

CS301 MID TERM BY IQRA AZHAR

CS301

Mid term

2023

Solved by

 **Iqra Azhar** 

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Virtual University Overseas Examination System V2.1 (x86)

Data Structures (CS301)

Question: 31 (Marks: 5) inviolator25 Attempted 0

Part a) What is the function of `length()` method in the queue?

Part b) Draw a complete binary tree containing five nodes.

Answer:

Part a) What is the function of `length()` method in the queue?

Solution:

The `length()` method in a queue returns the total number of elements currently present in the queue.

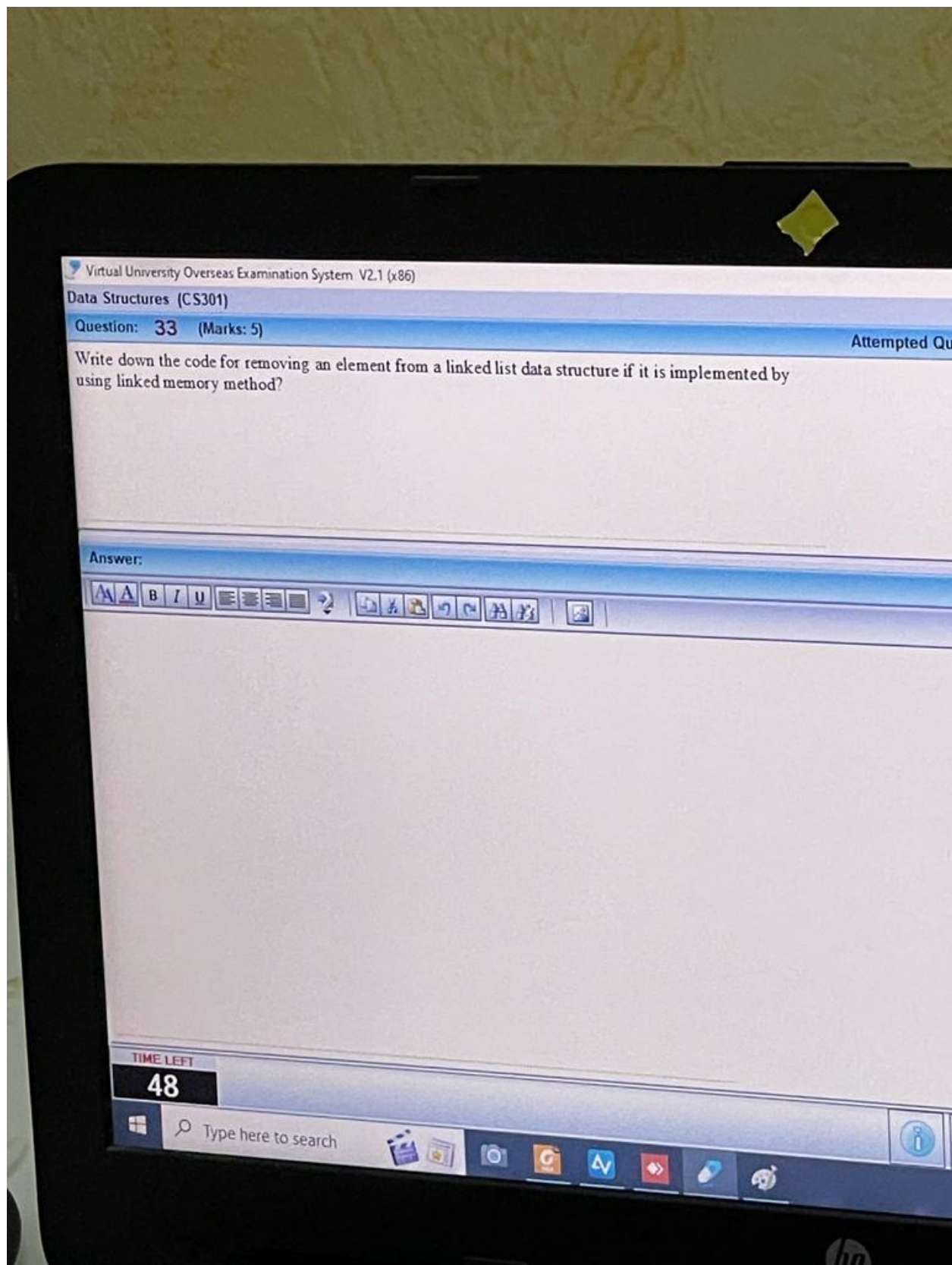
Part b) Draw a complete binary tree containing five nodes.

Solution:

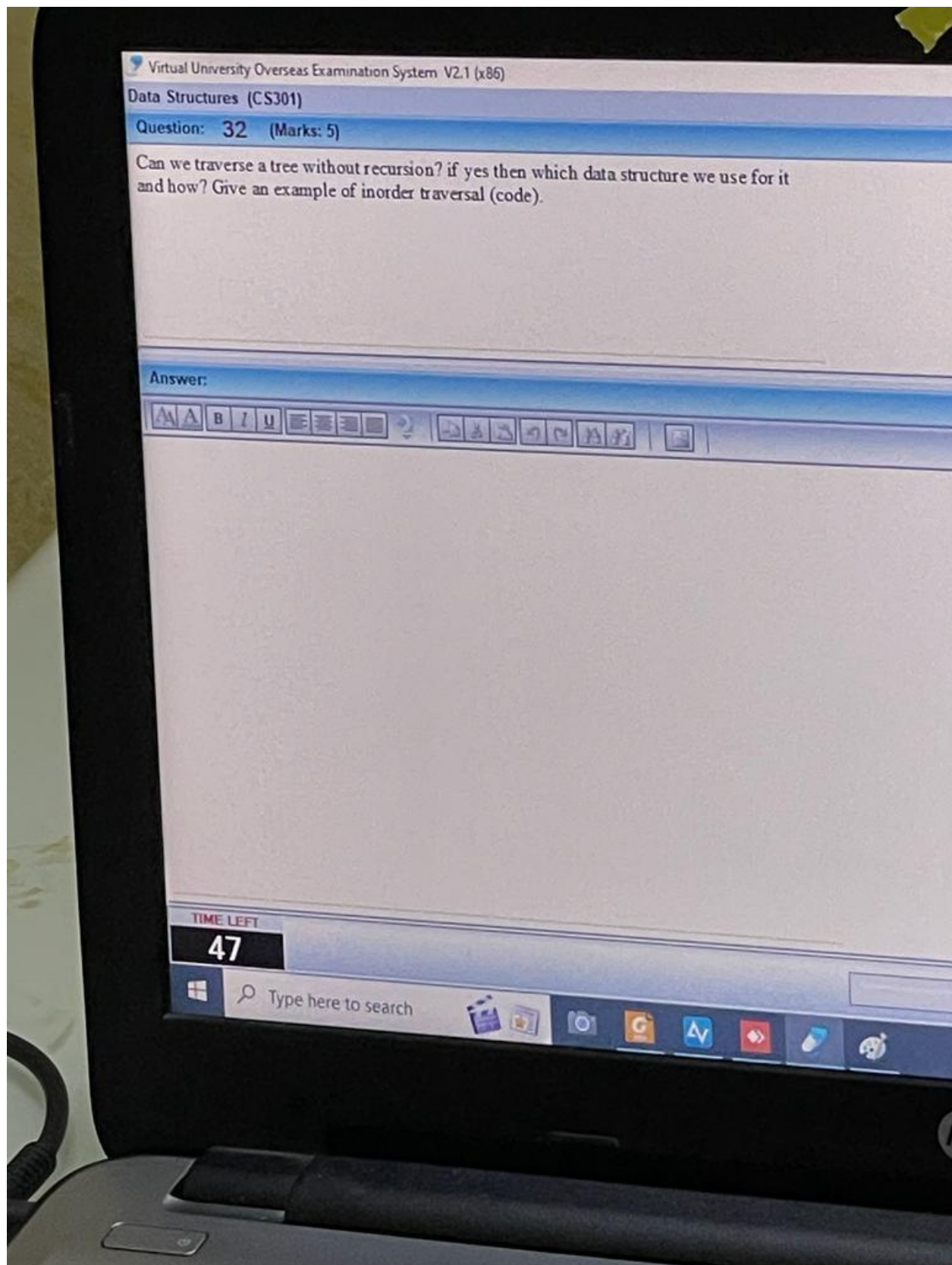
```
graph TD; A --> B; A --> C; B --> D; B --> E
```

TIME LEFT 37

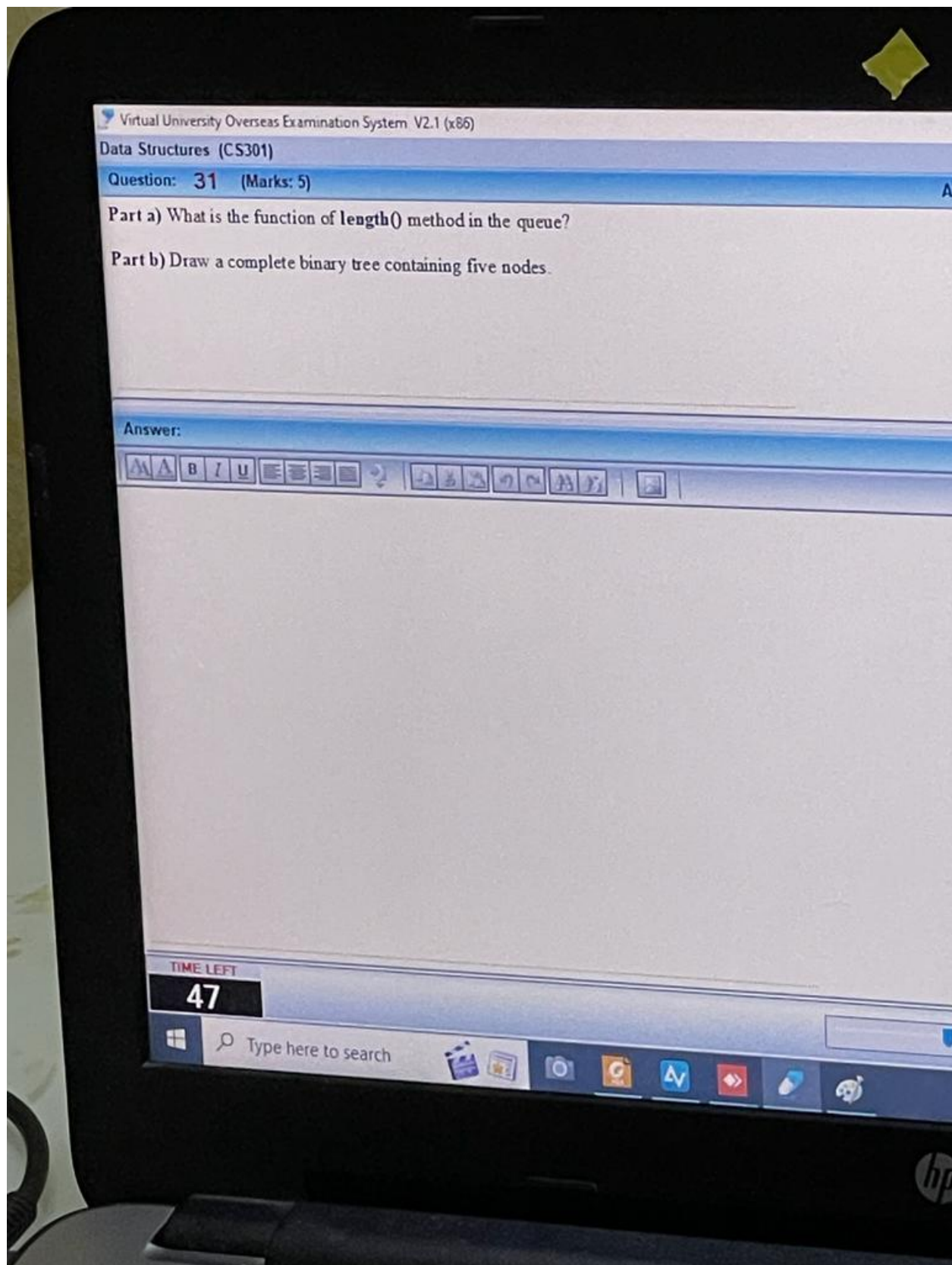
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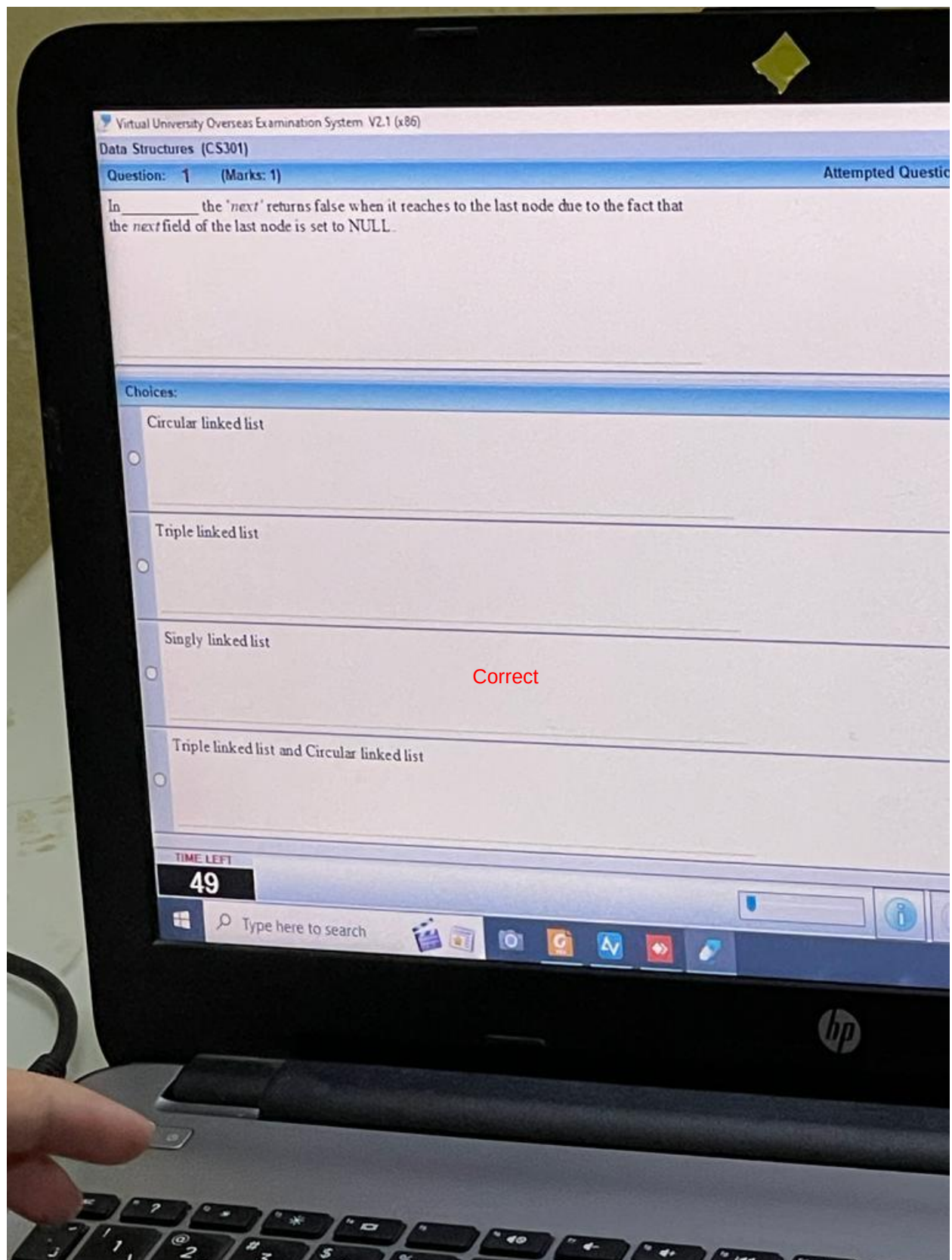
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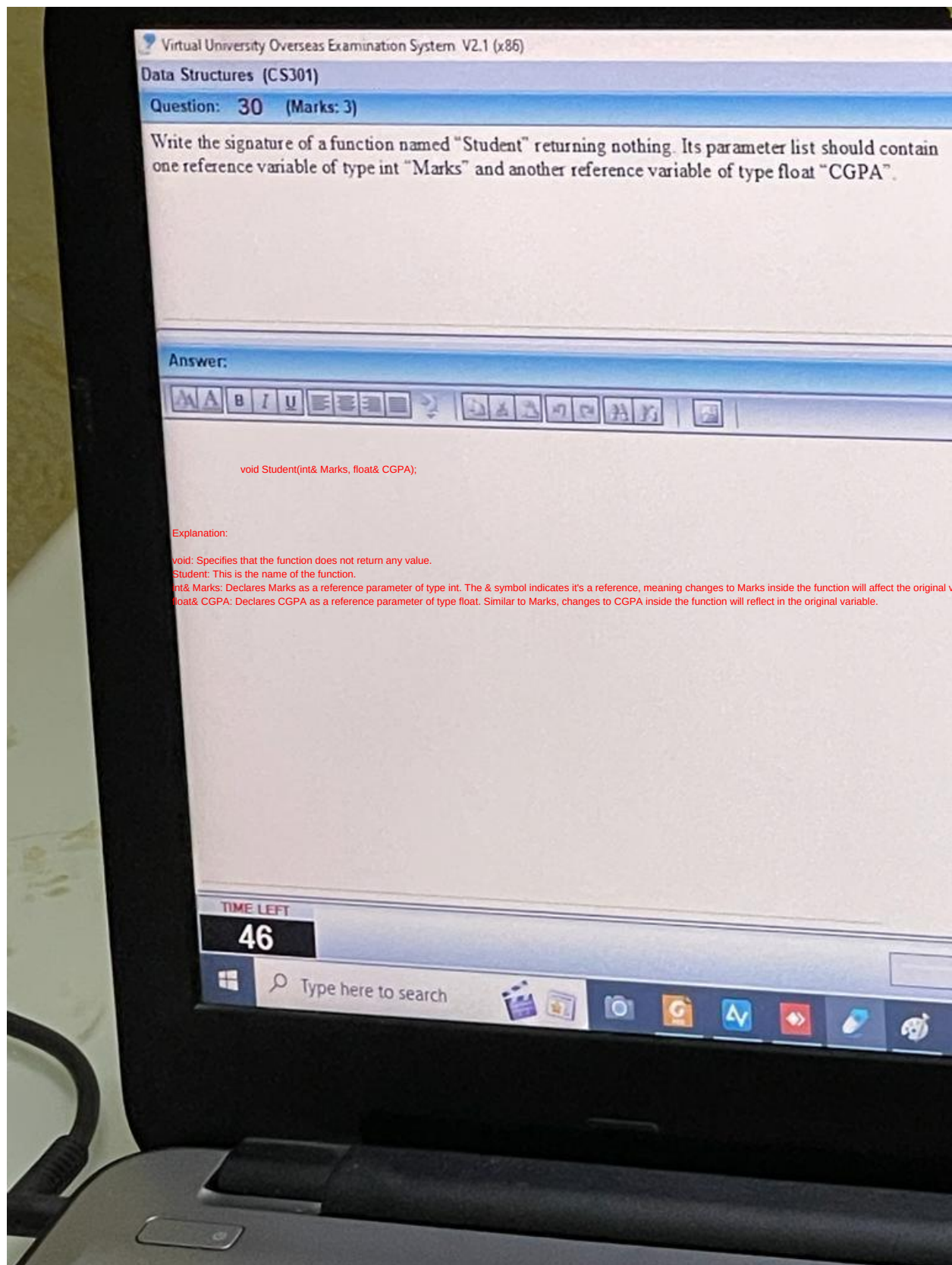
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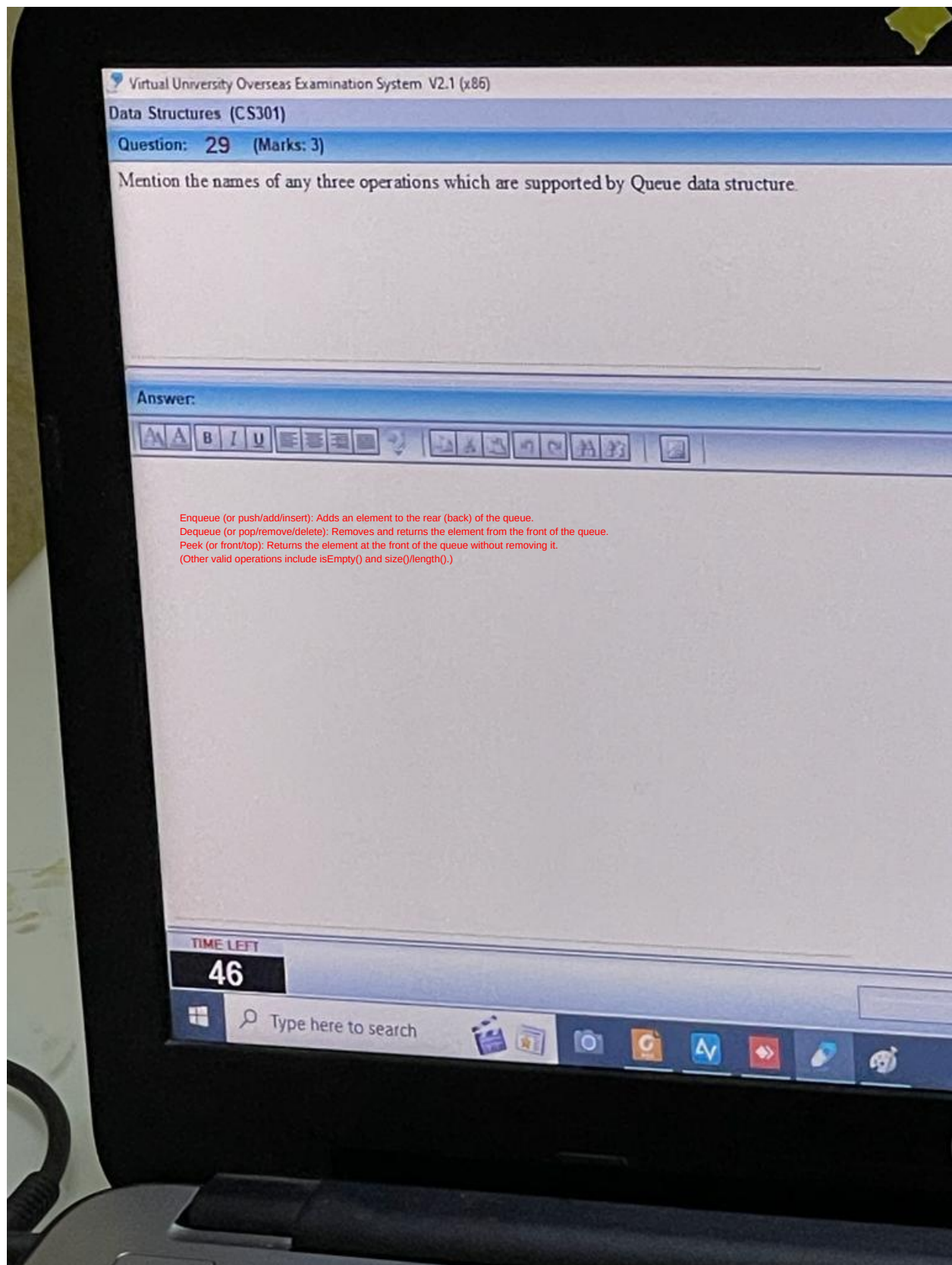
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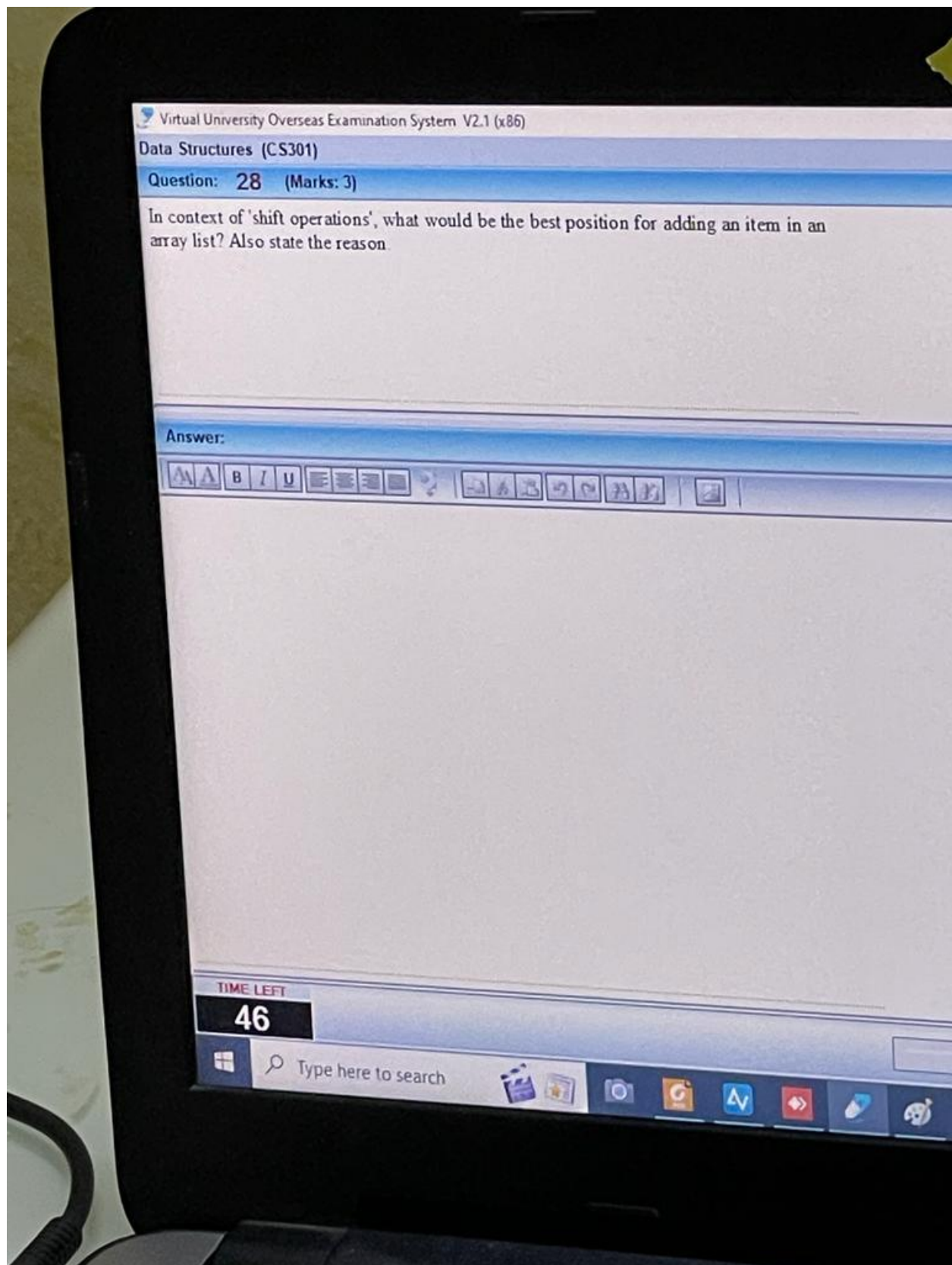
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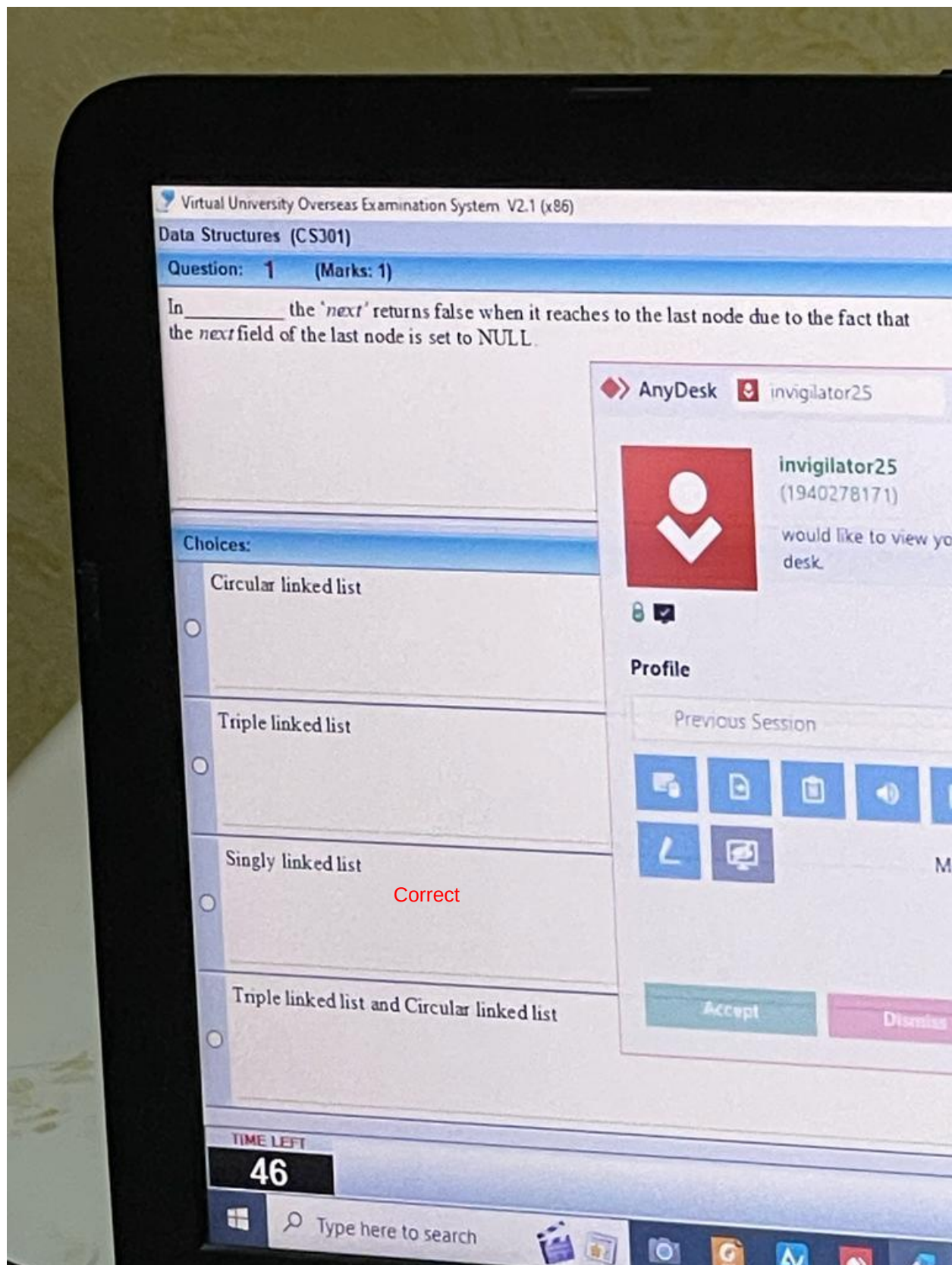
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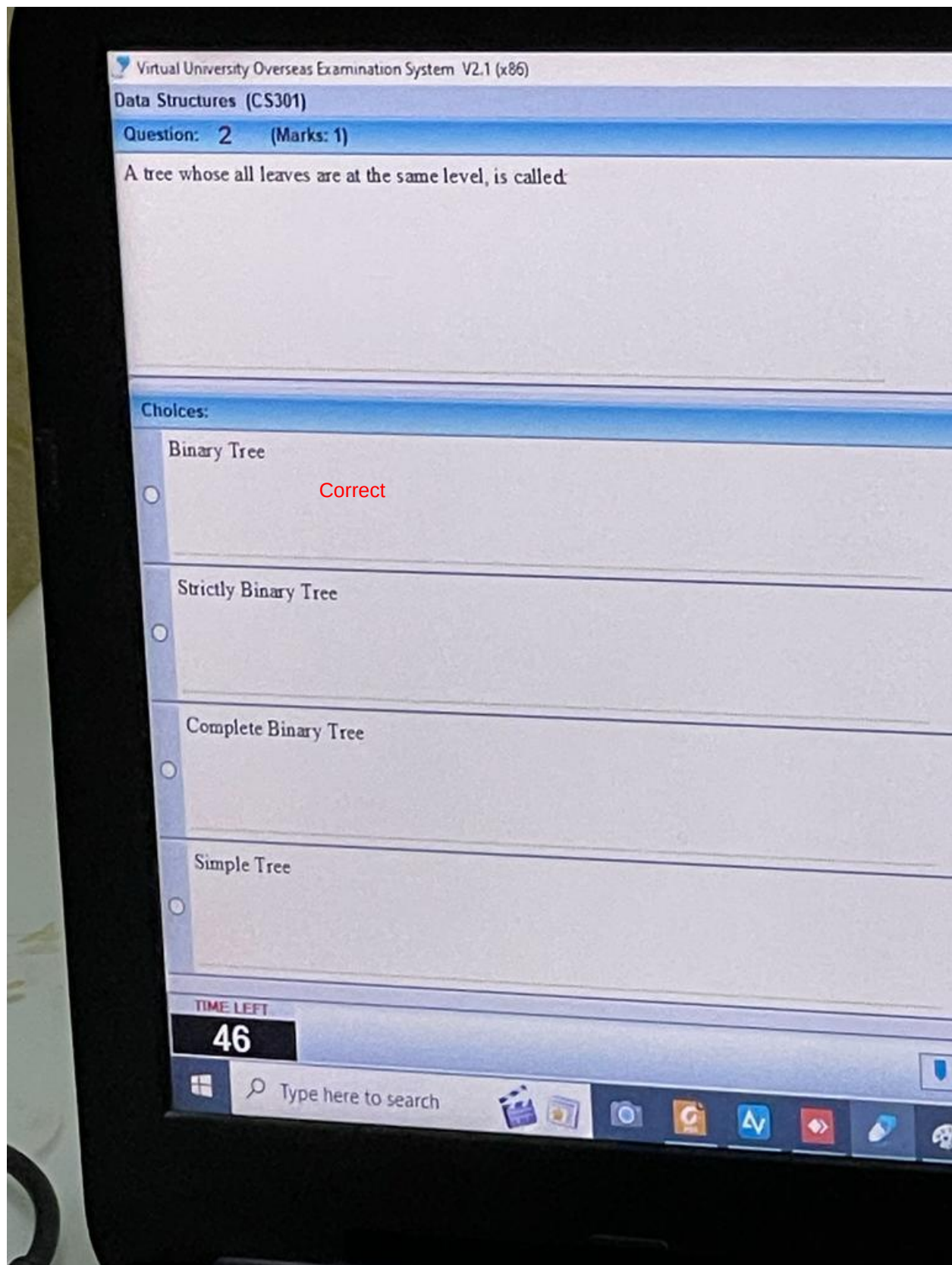
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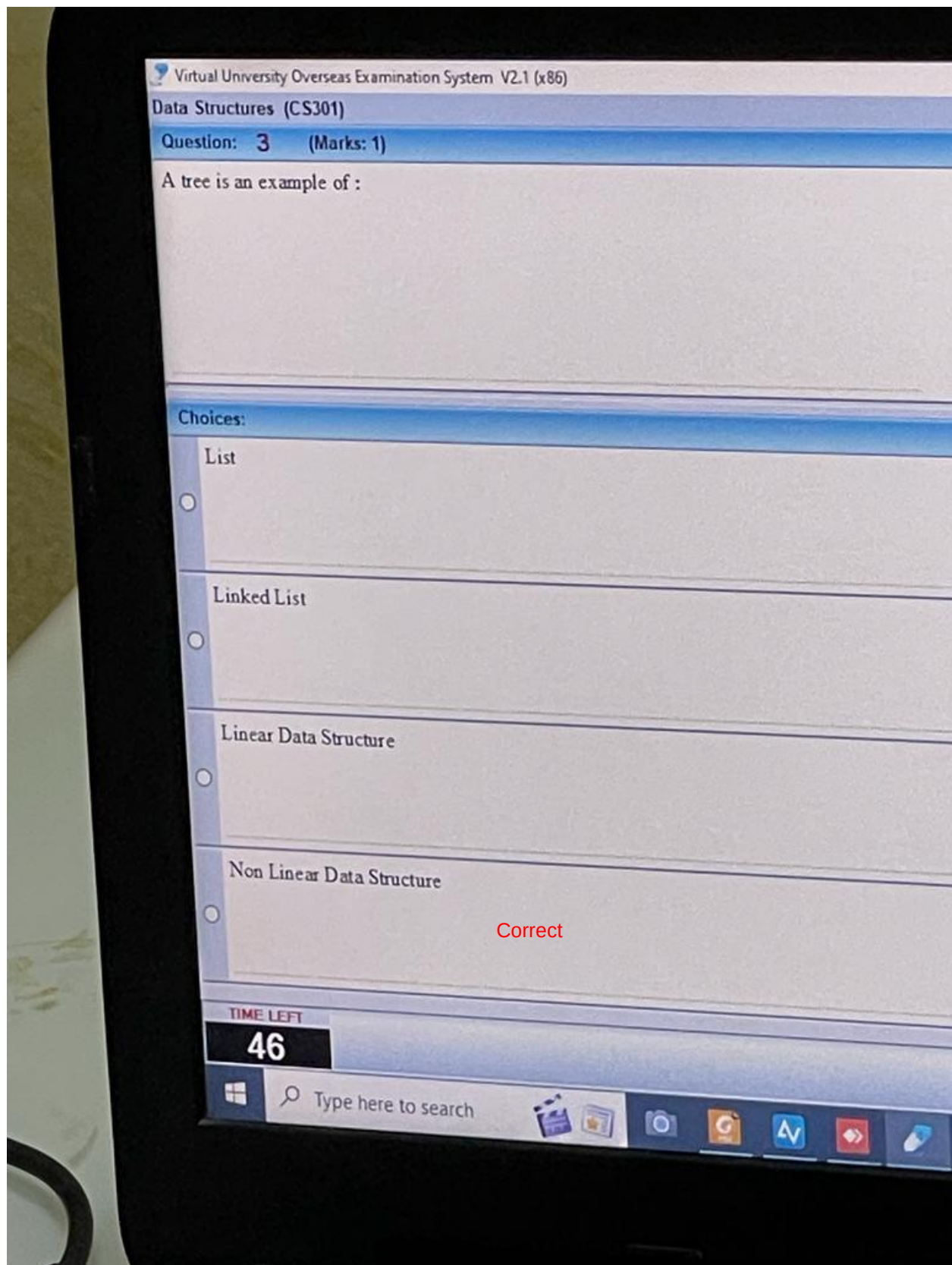
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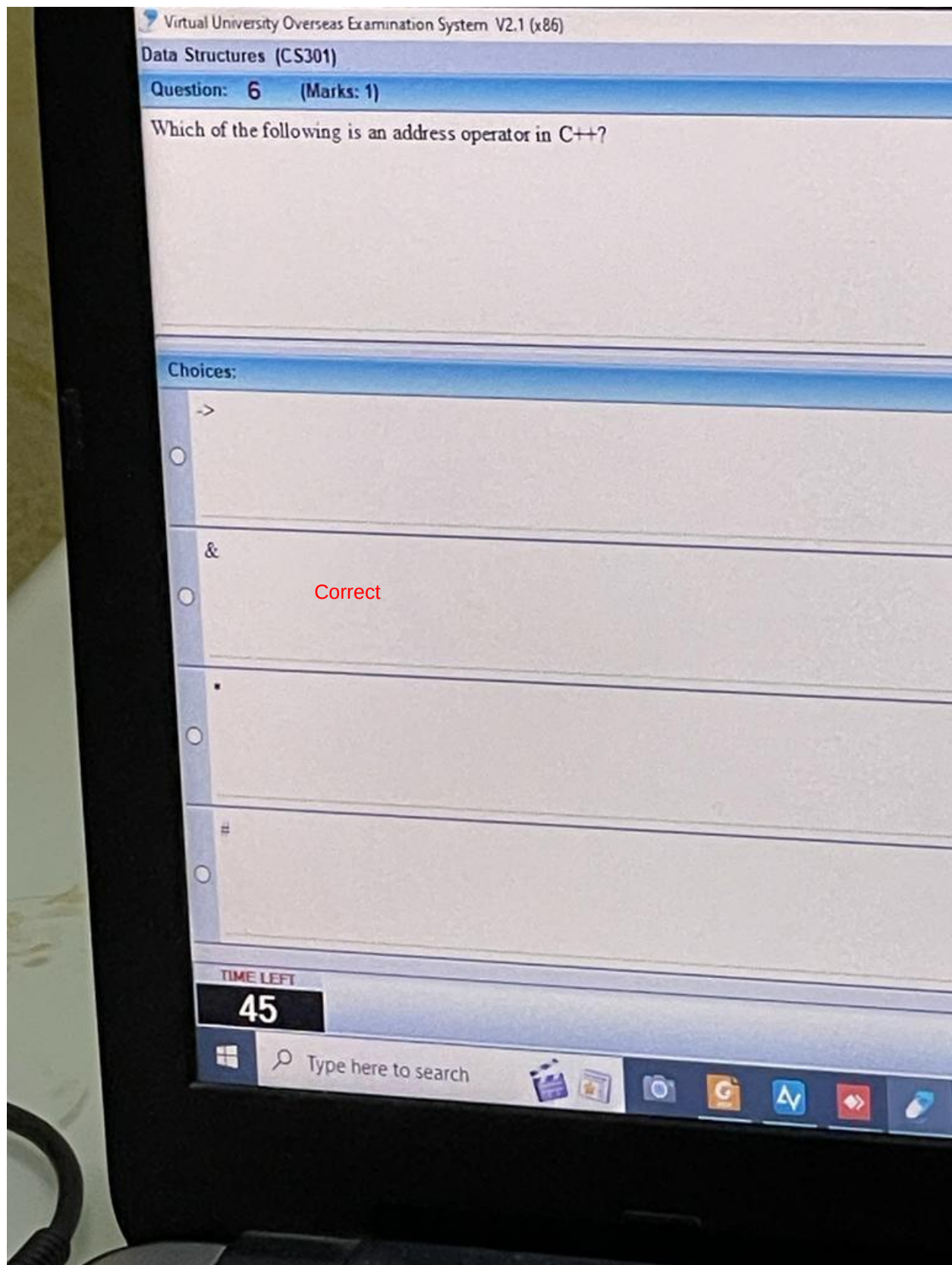
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Virtual University Overseas Examination System V2.1 (x86)

Data Structures (CS301)

Question: 5 (Marks: 1)

Which of the following statement is NOT true for reference variable?

Choices:

☐ Once a reference is created, it cannot be later made to reference another object.

☐ References cannot be NULL

☒ References can be uninitialized.

☐ It is not possible to refer directly to a reference object after it is defined

Correct

TIME LEFT

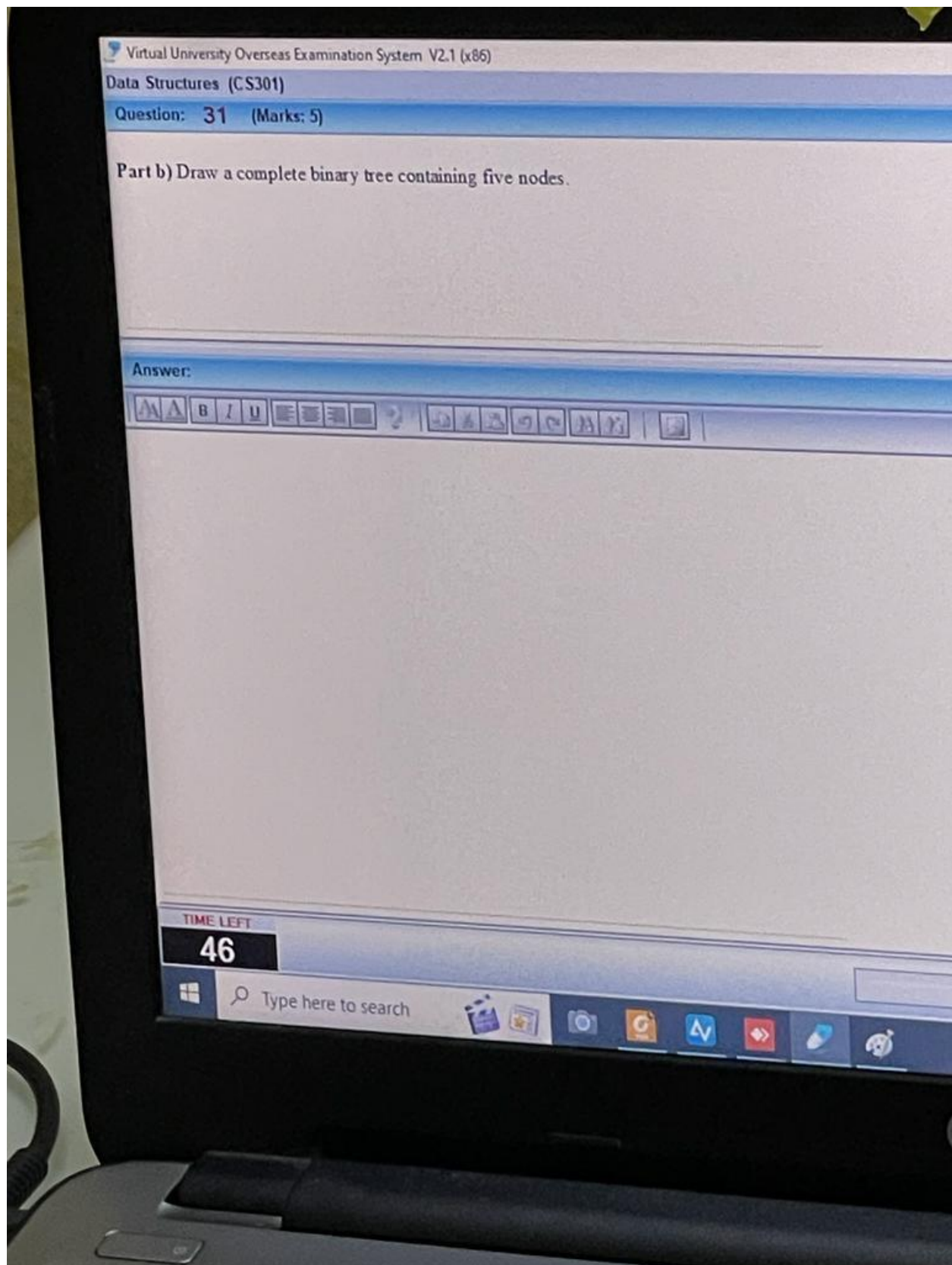
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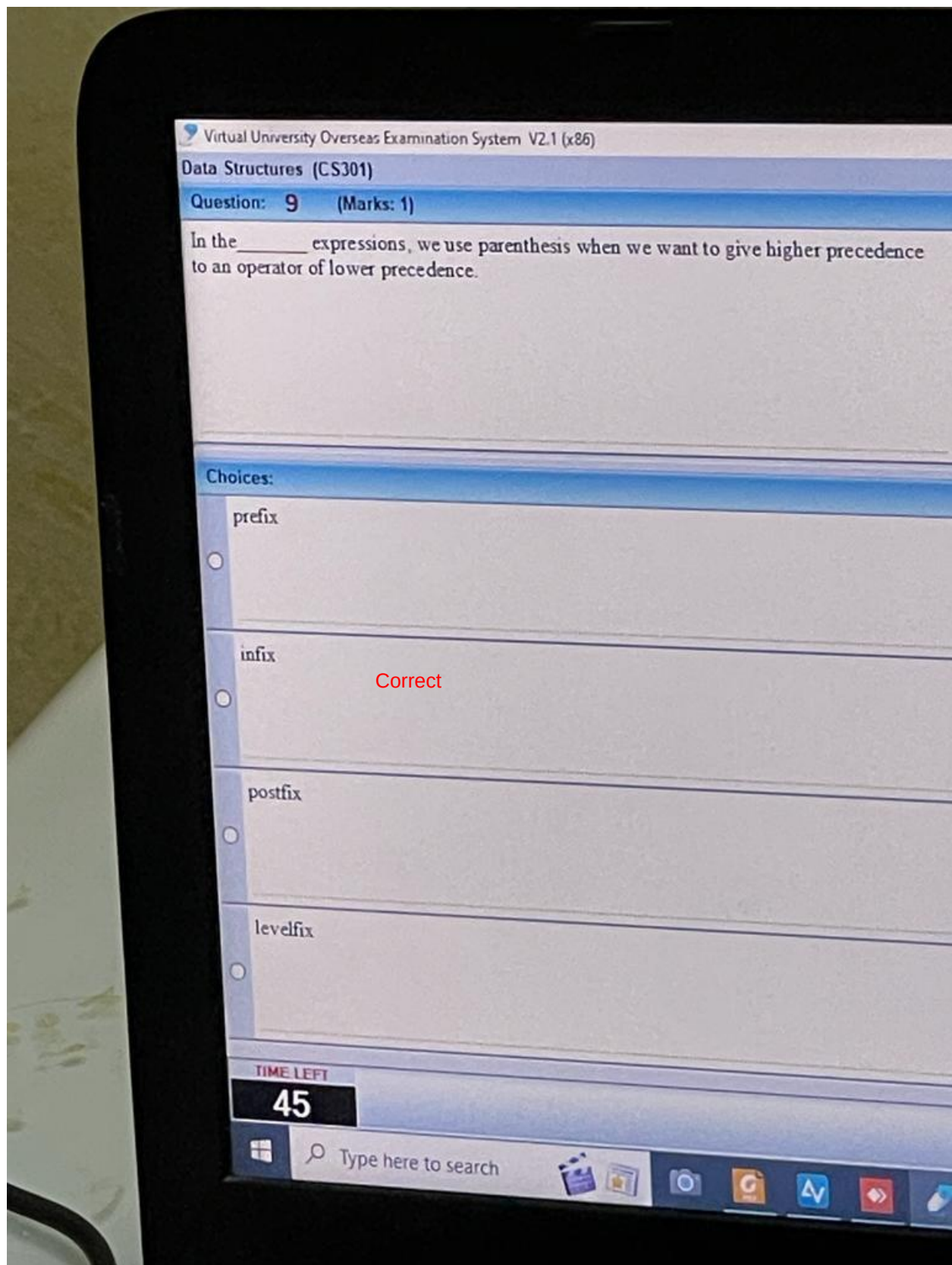
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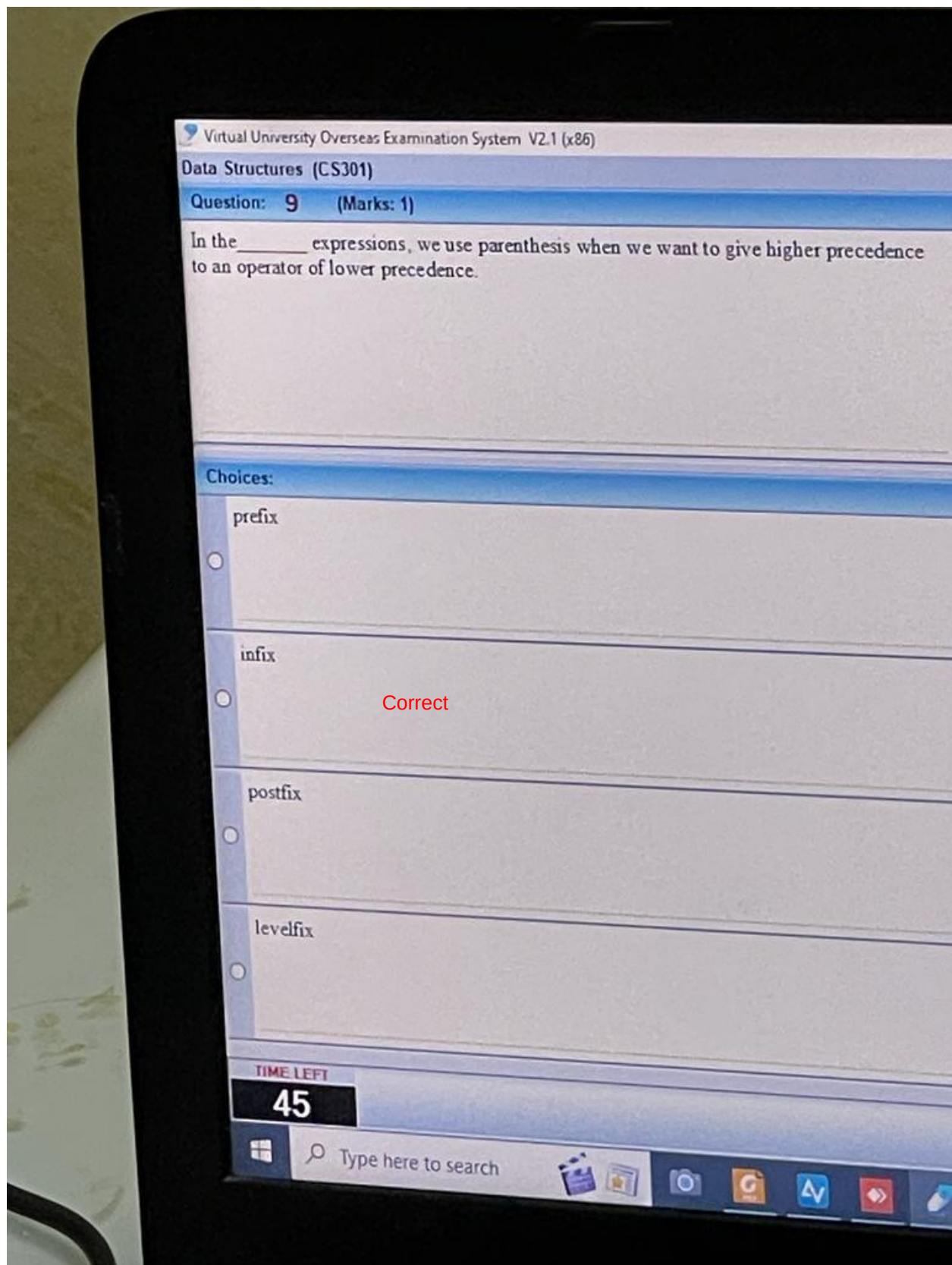
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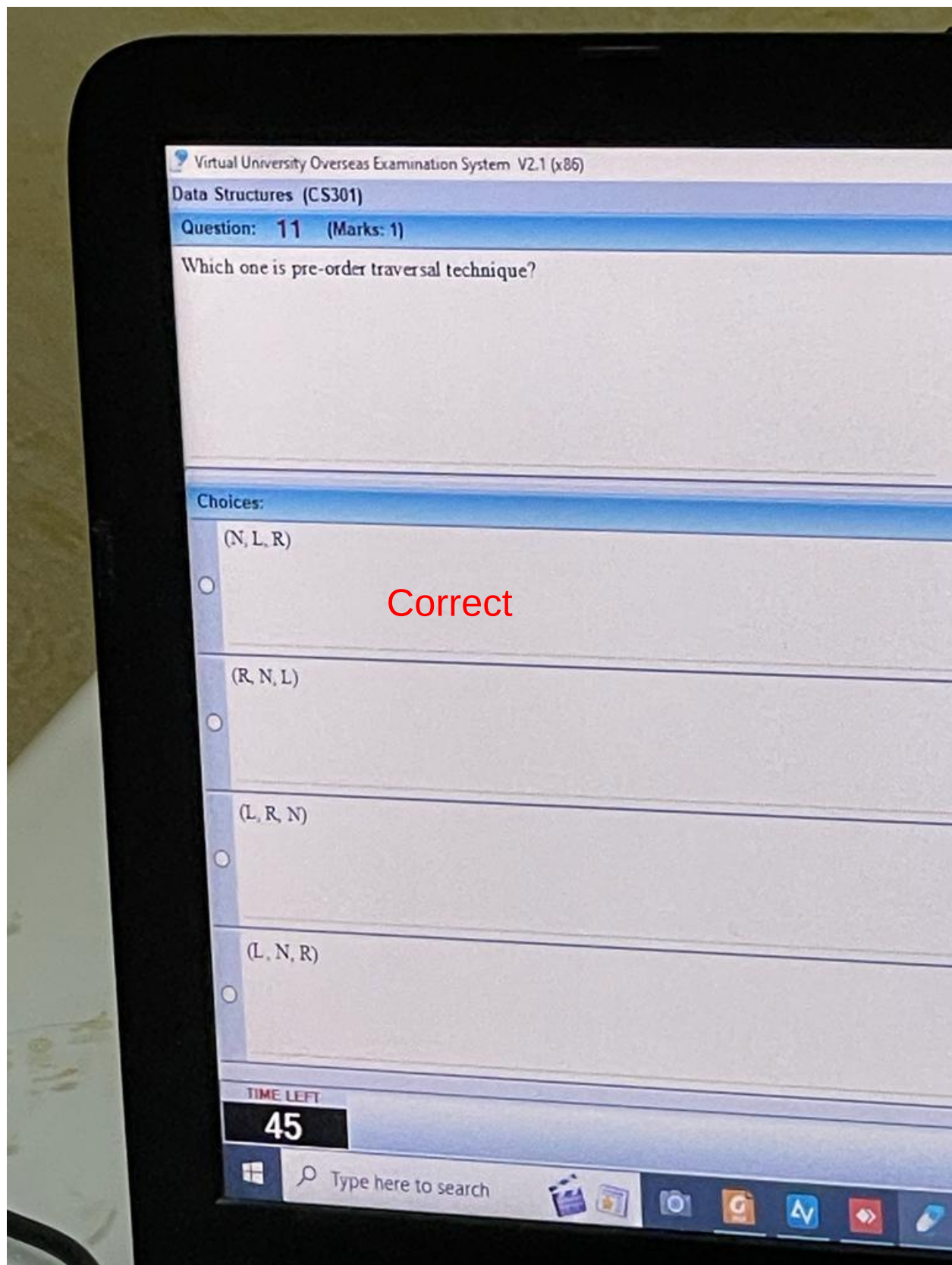
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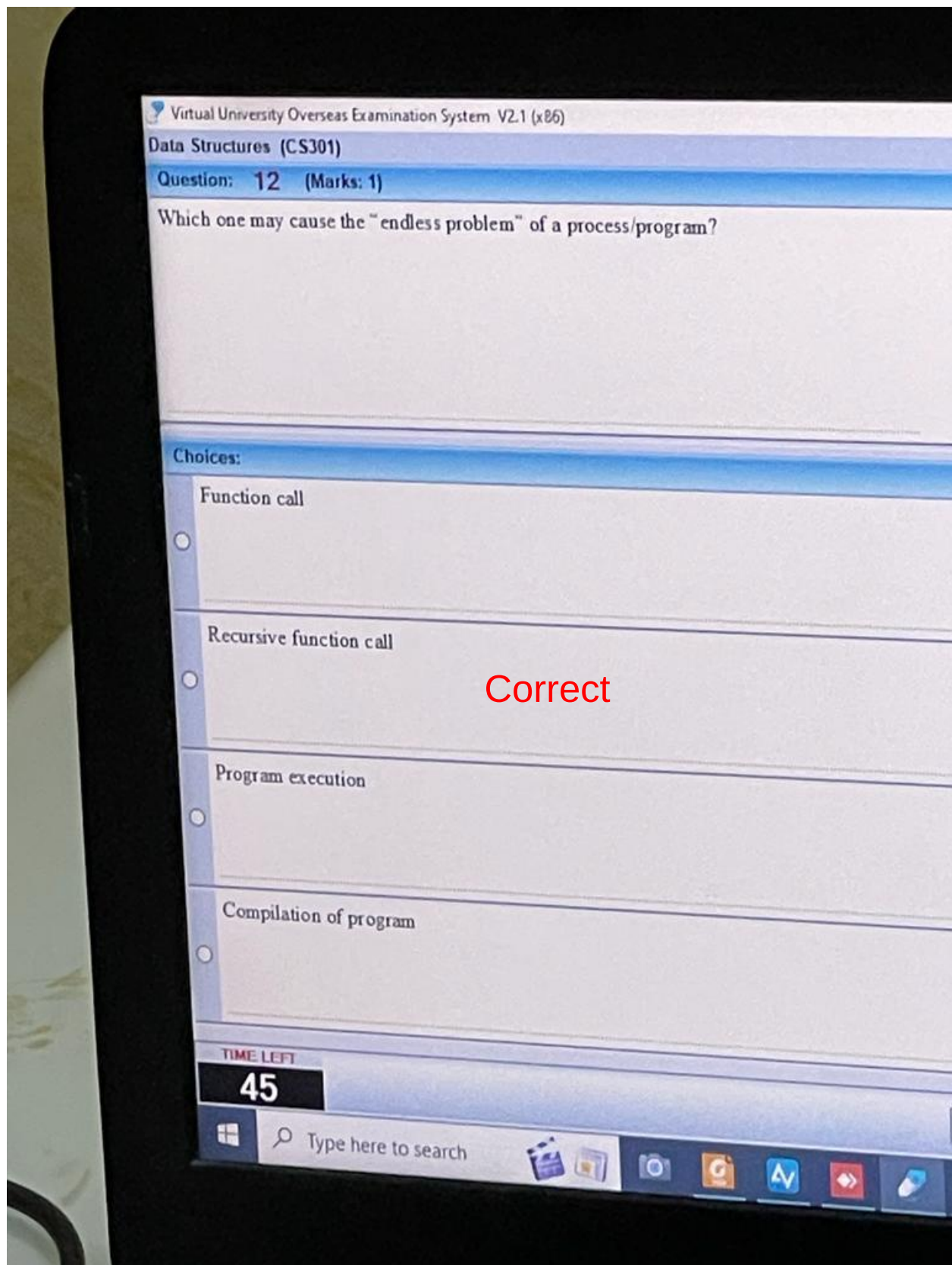
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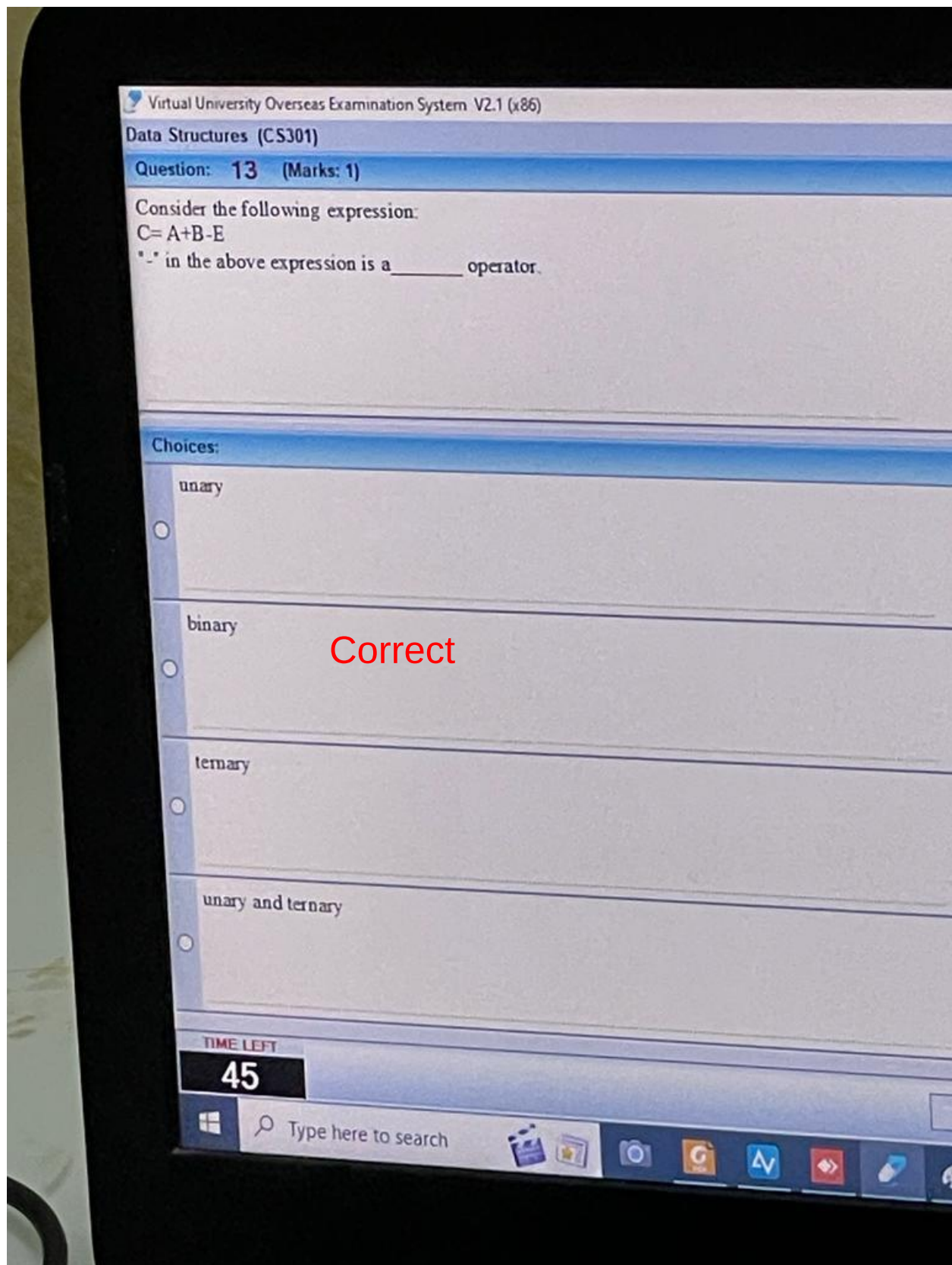
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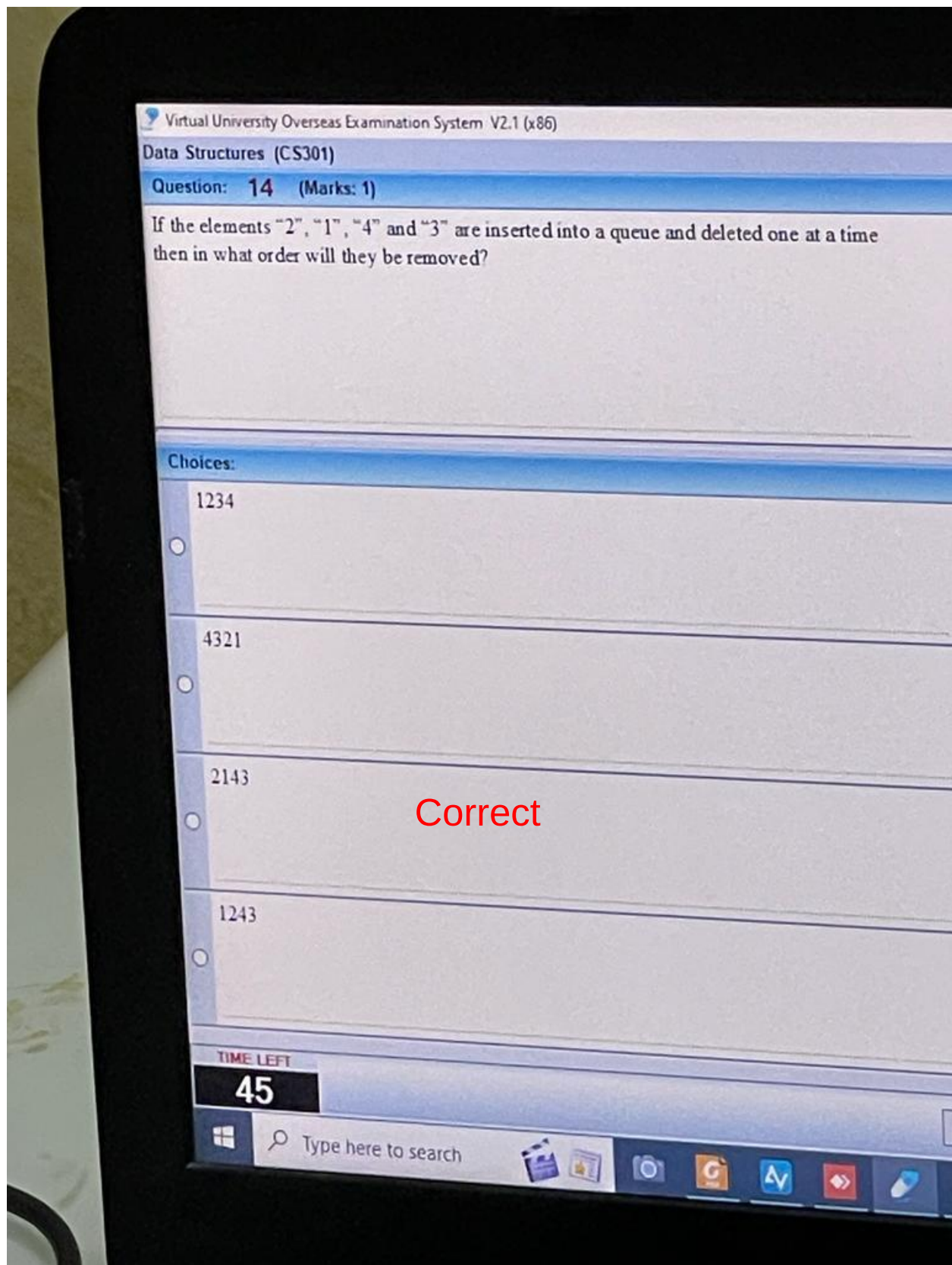
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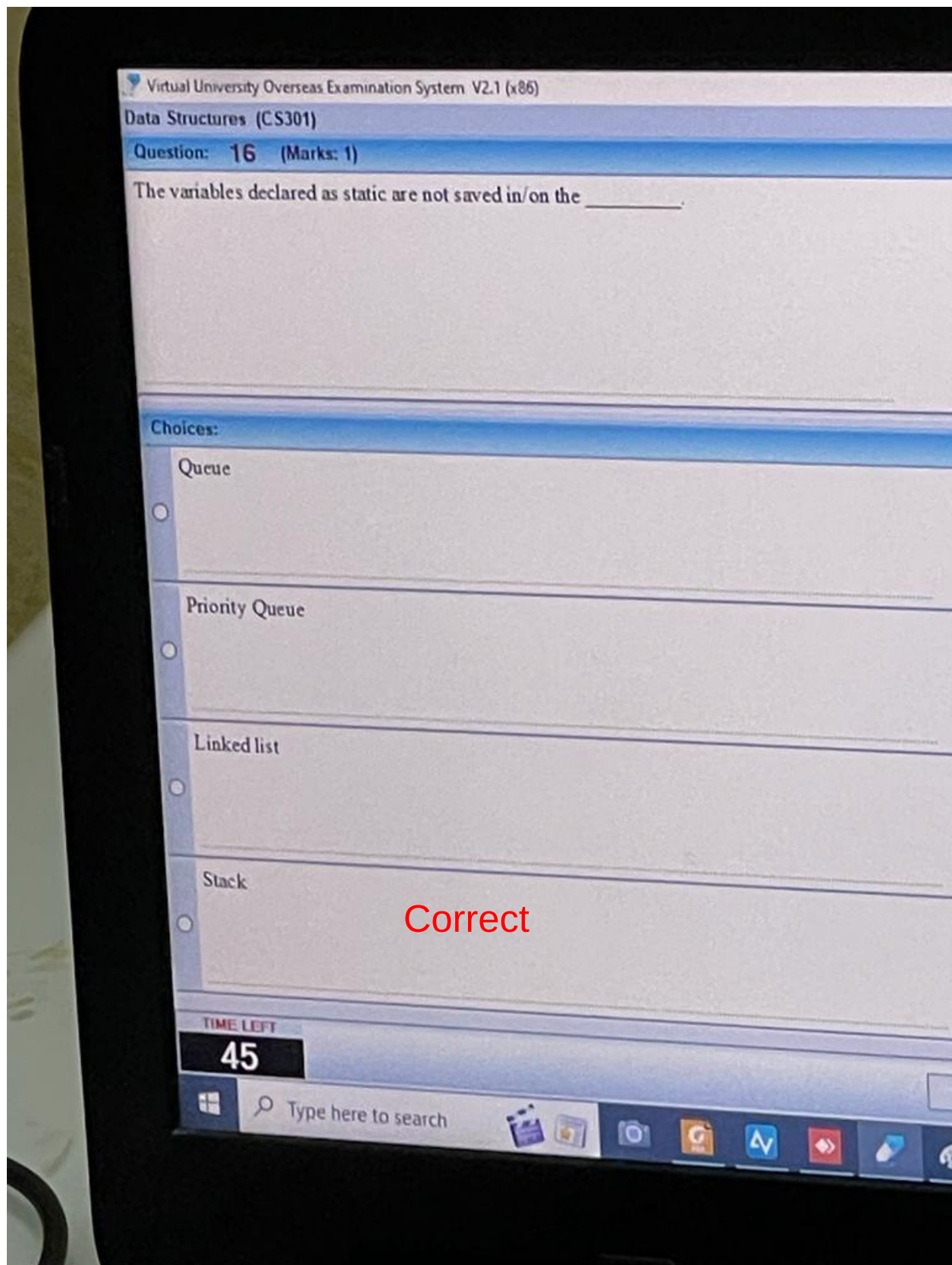
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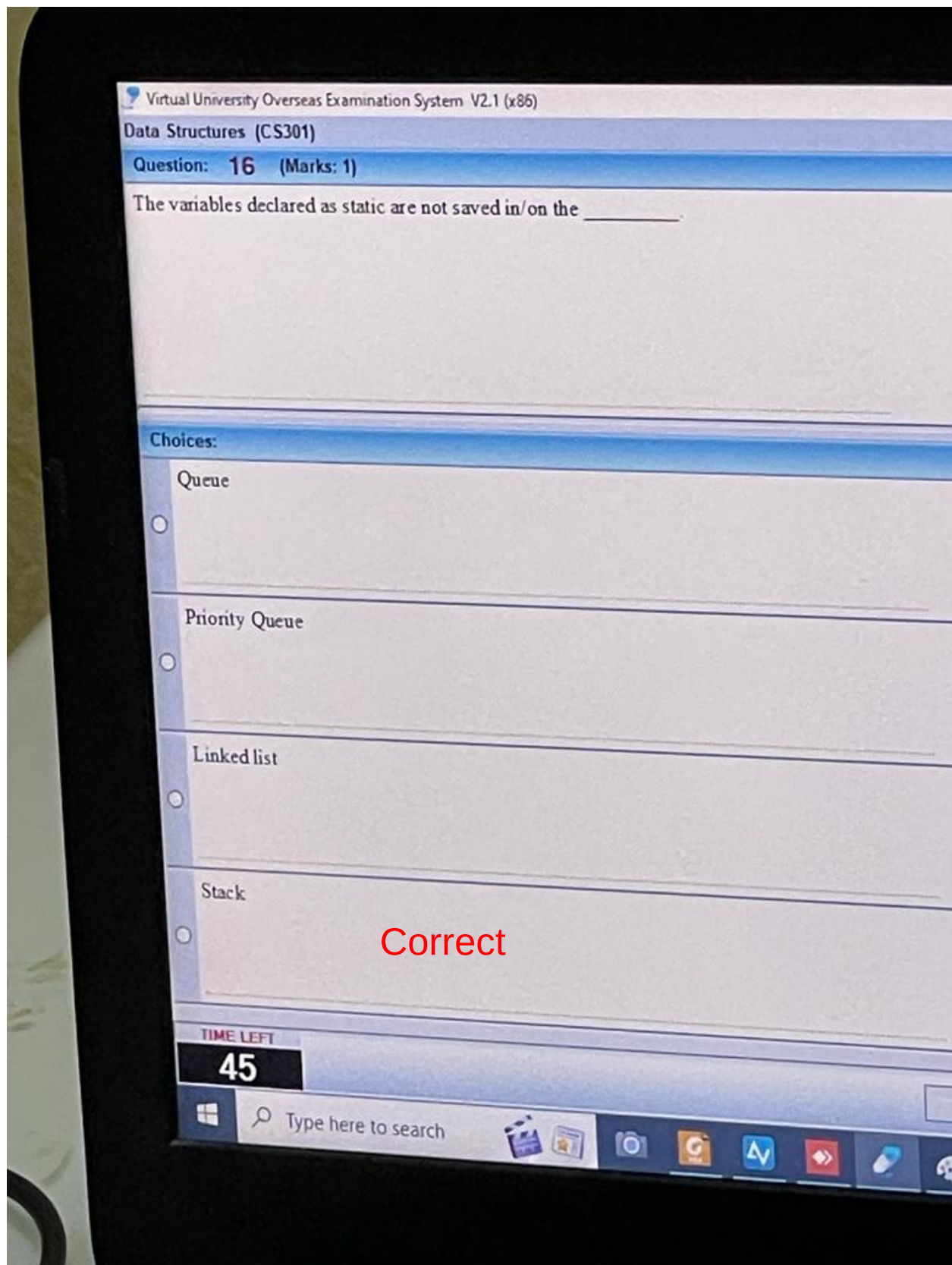
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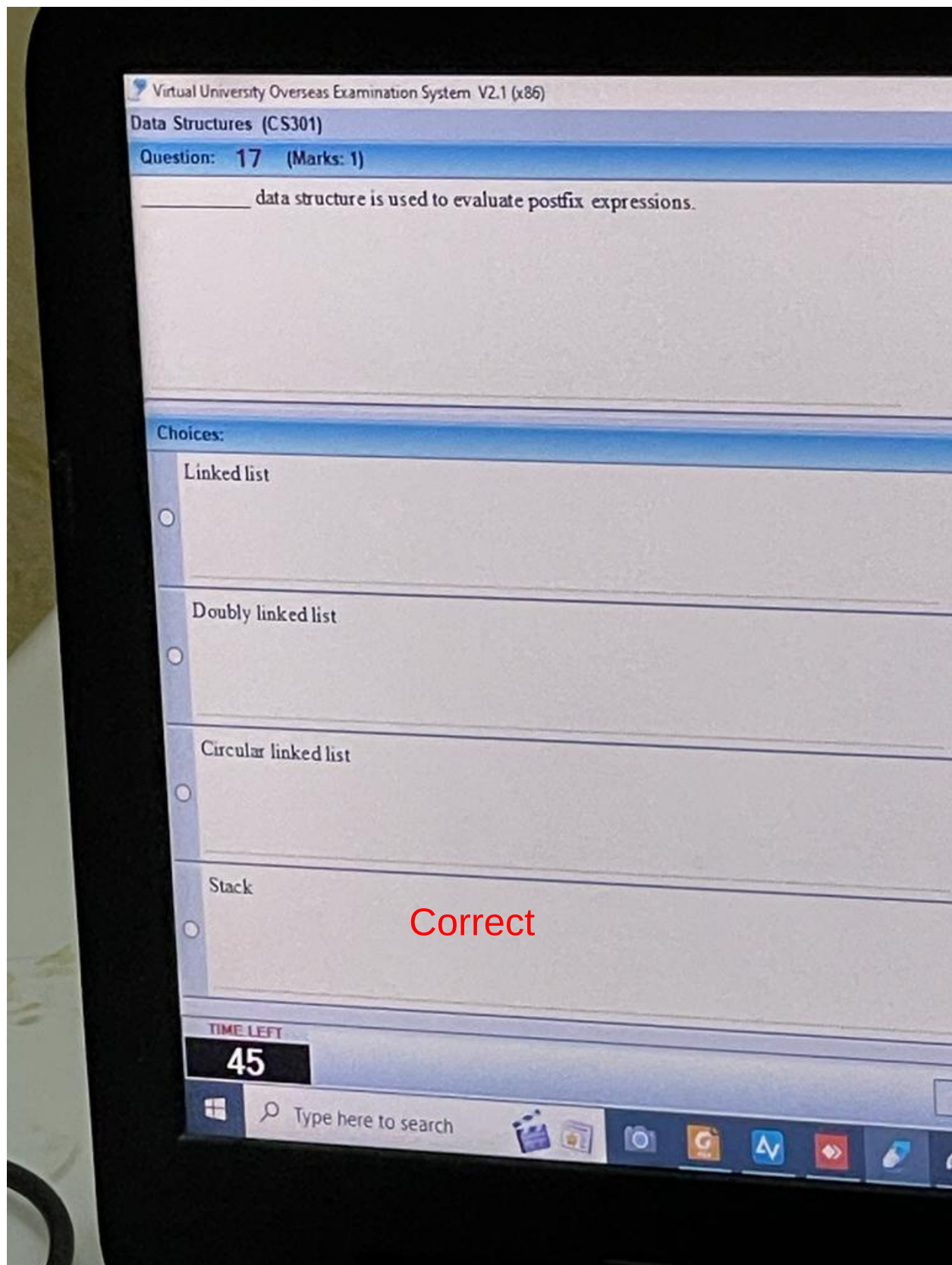
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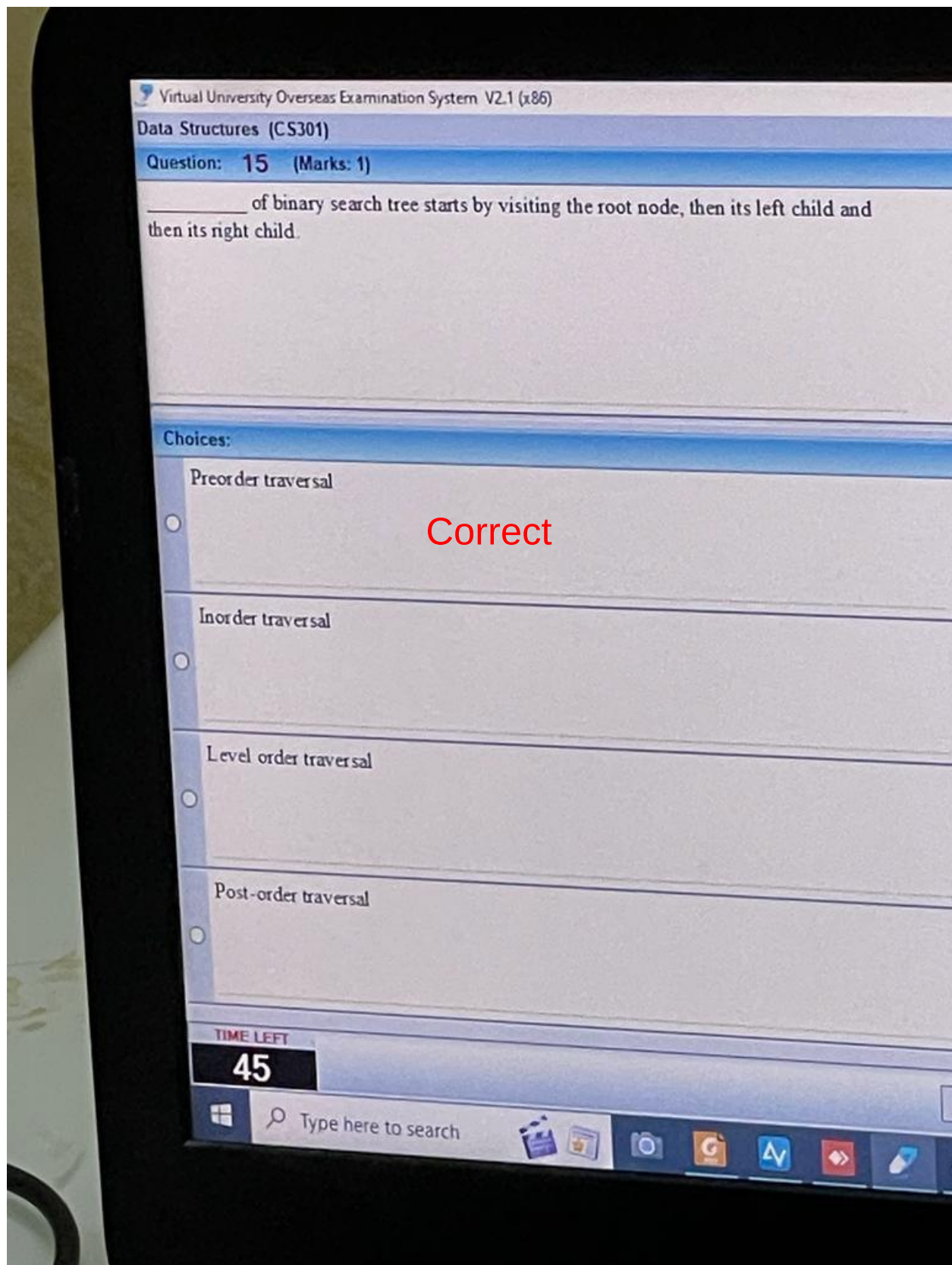
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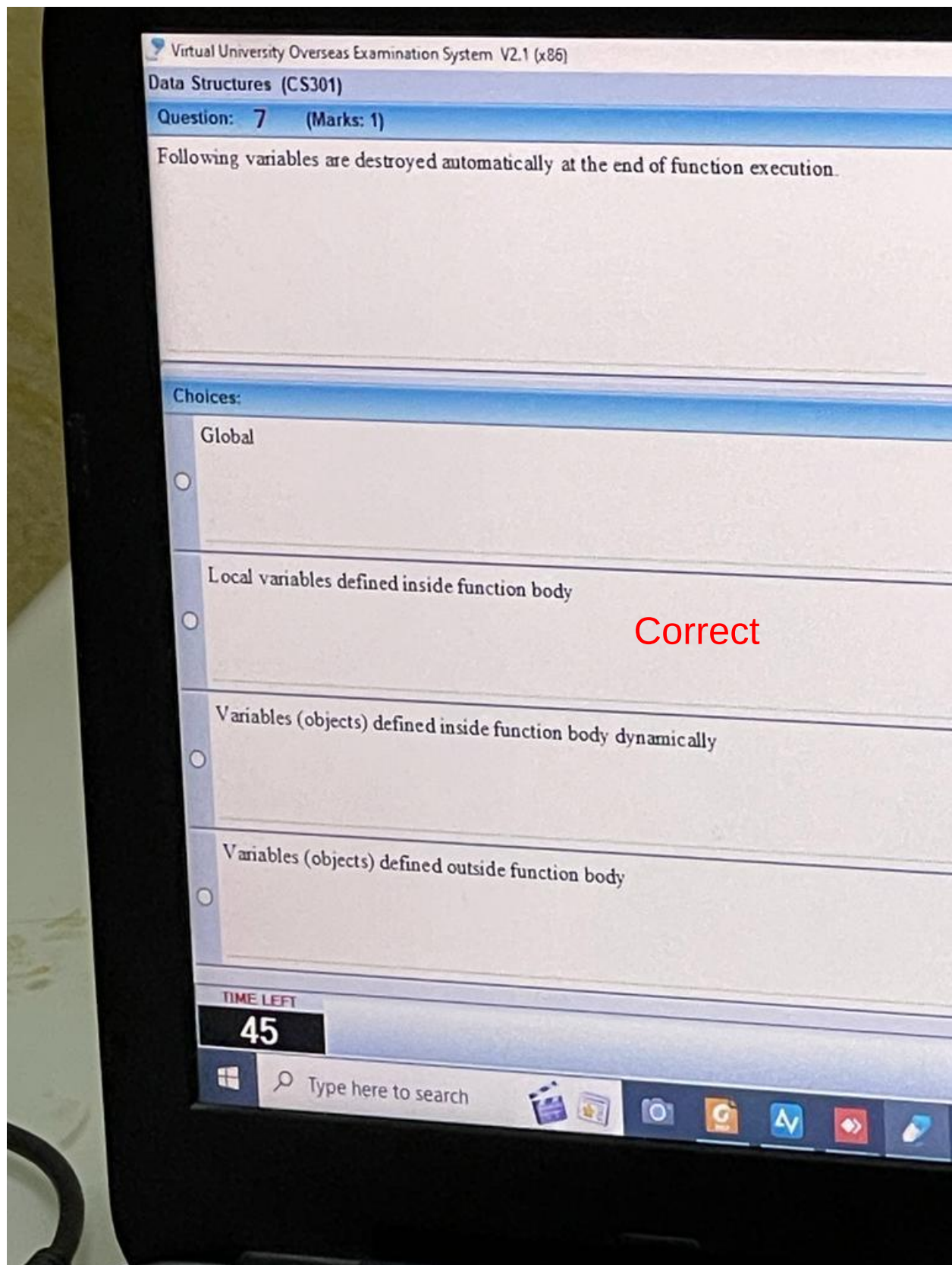
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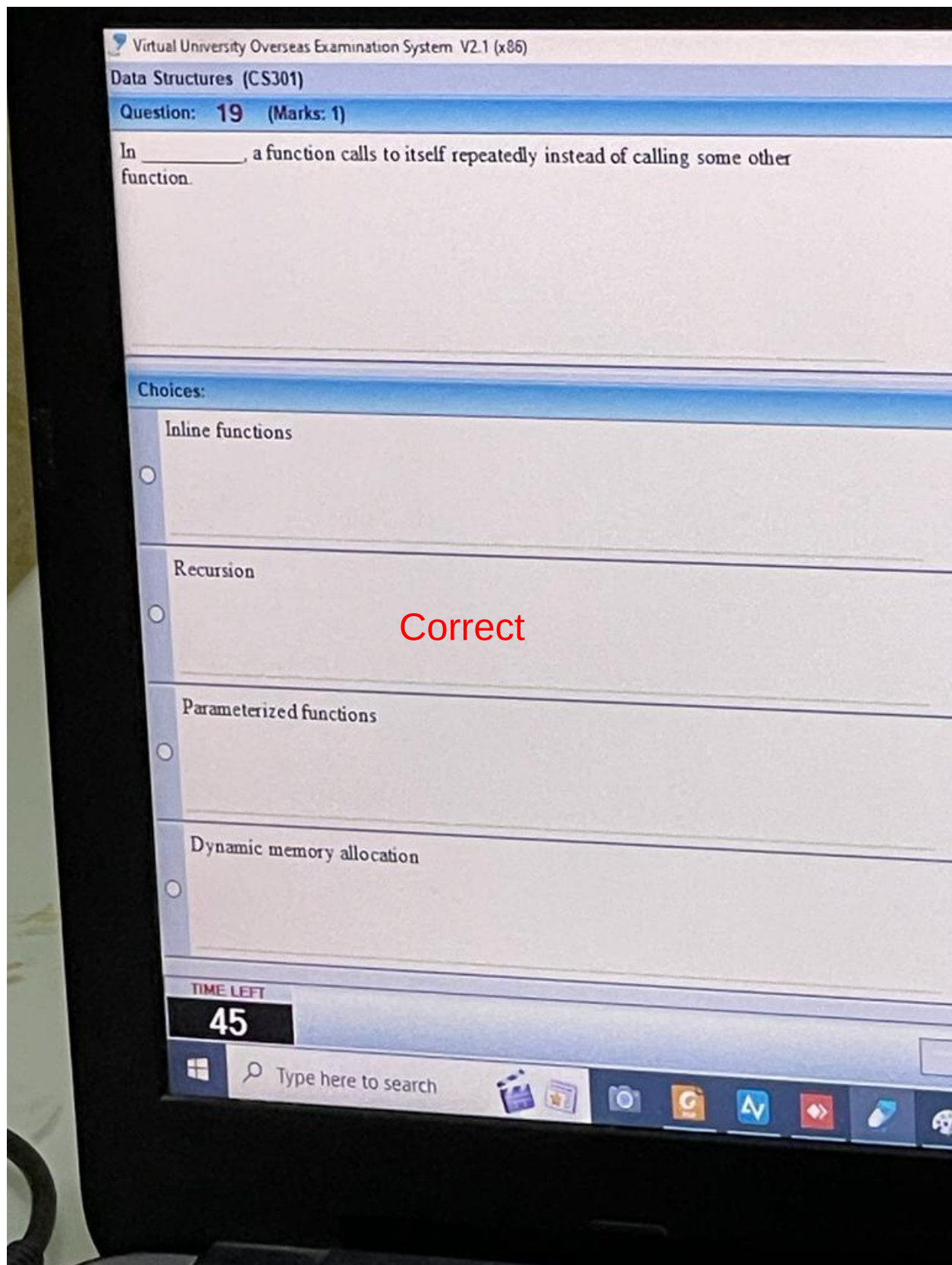
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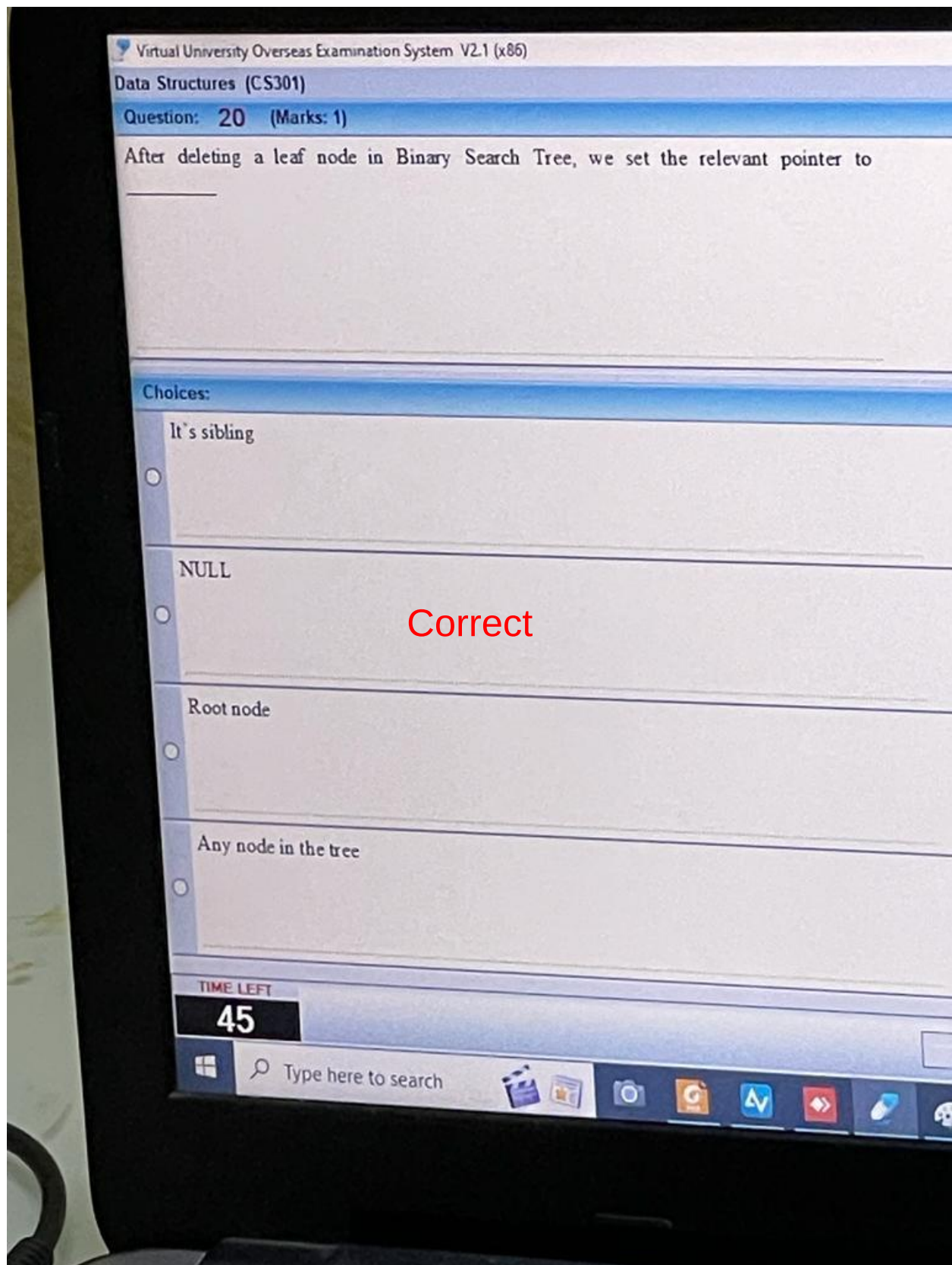
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Virtual University Overseas Examination System V2.1 (x86)

Data Structures (CS301)

Question: 21 (Marks: 1)

In finding minimum node from Binary Search Tree, we first check if _____

Choices:

☐ Tree has maximum value

☒ Tree is NULL

☐ Left most leaf node is present

☐ Right most leaf node is present

Correct

TIME LEFT

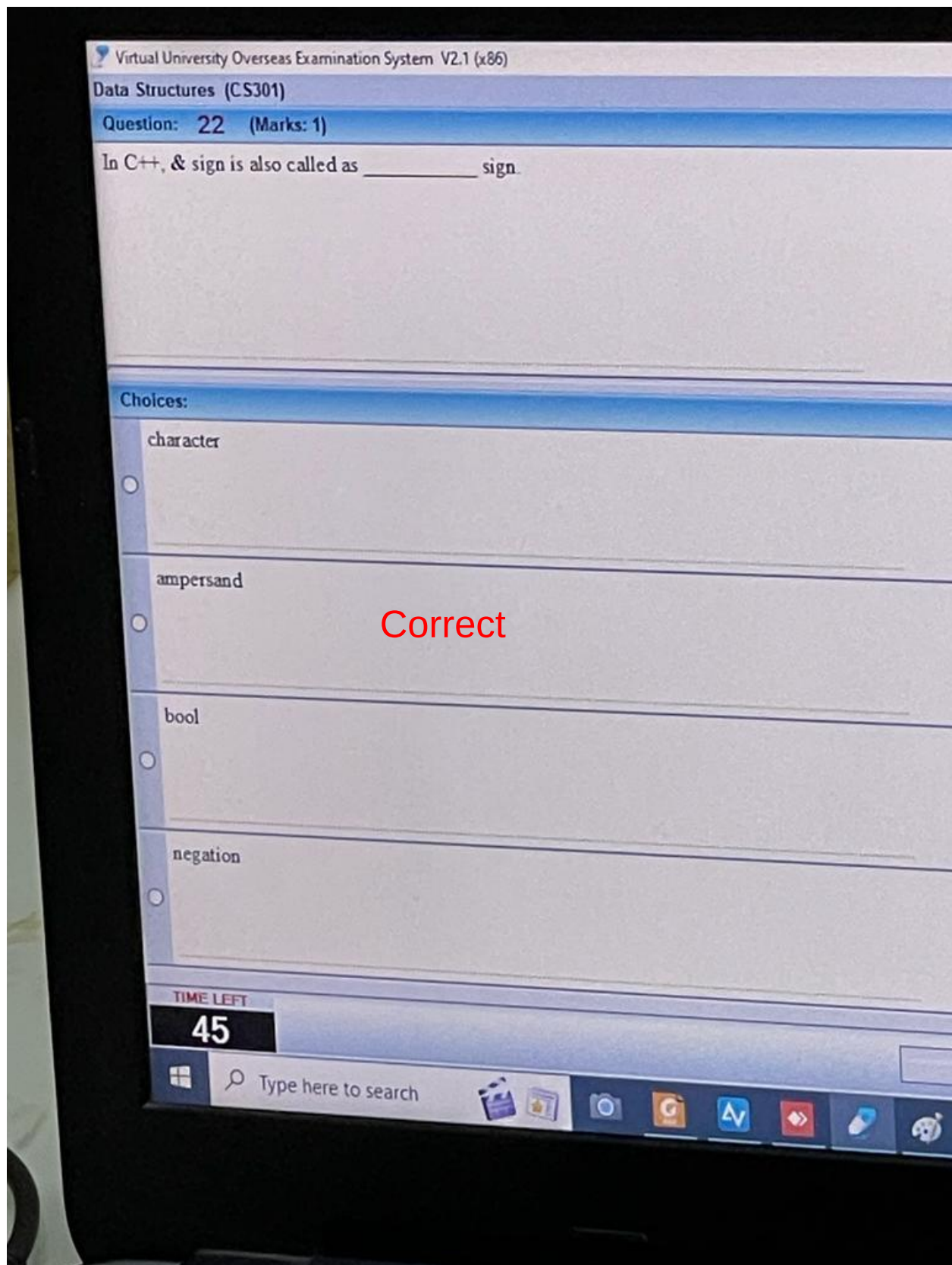
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Virtual University Overseas Examination System V2.1 (x86)

Data Structures (CS301)

Question: 24 (Marks: 1)

While deleting a node from Binary Search Tree (BST), there is/are _____ number of case(s) we have to take care of.

Choices:

one

☐

two

☐

three

☐

four

☐

Correct

TIME LEFT

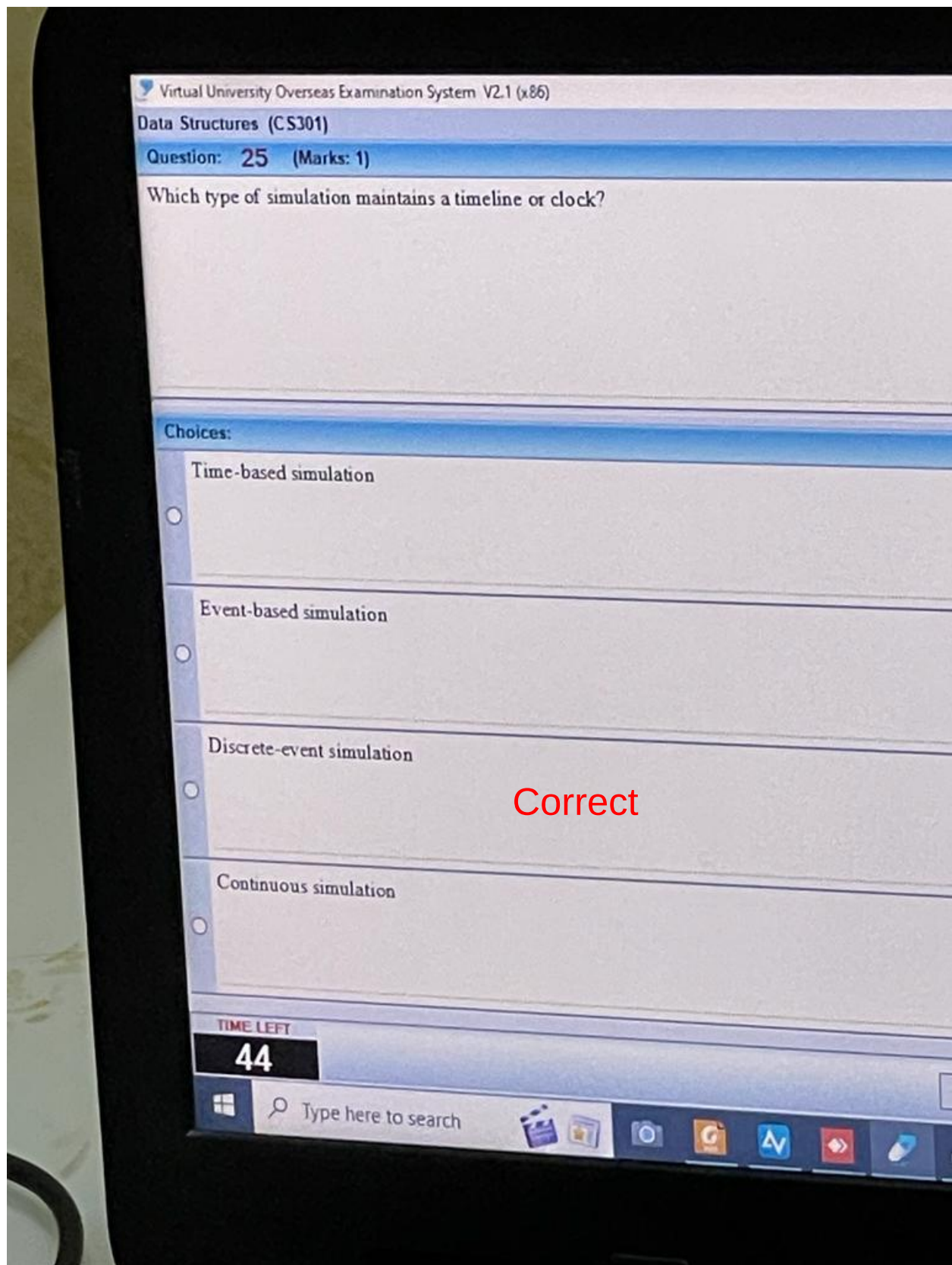
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Virtual University Overseas Examination System V2.1 (x86)

Data Structures (CS301)

Question: 26 (Marks: 1)

_____ is the variation of queue that does not follow FIFO rule.

Choices:

☐ Priority Queue

Correct

☐ Linear Queue

☐ Non-Linear Queue

☐ Circular Queue

TIME LEFT

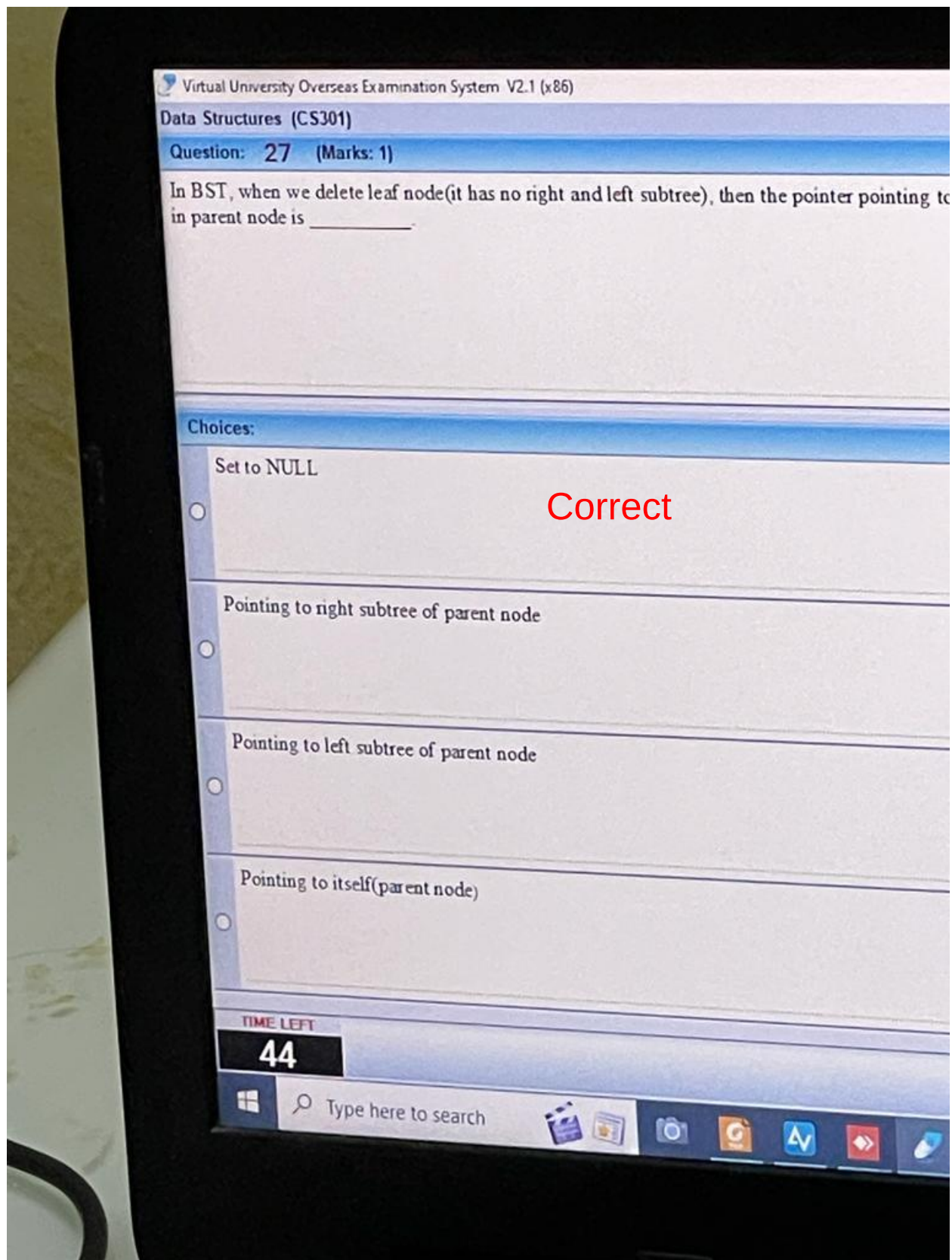
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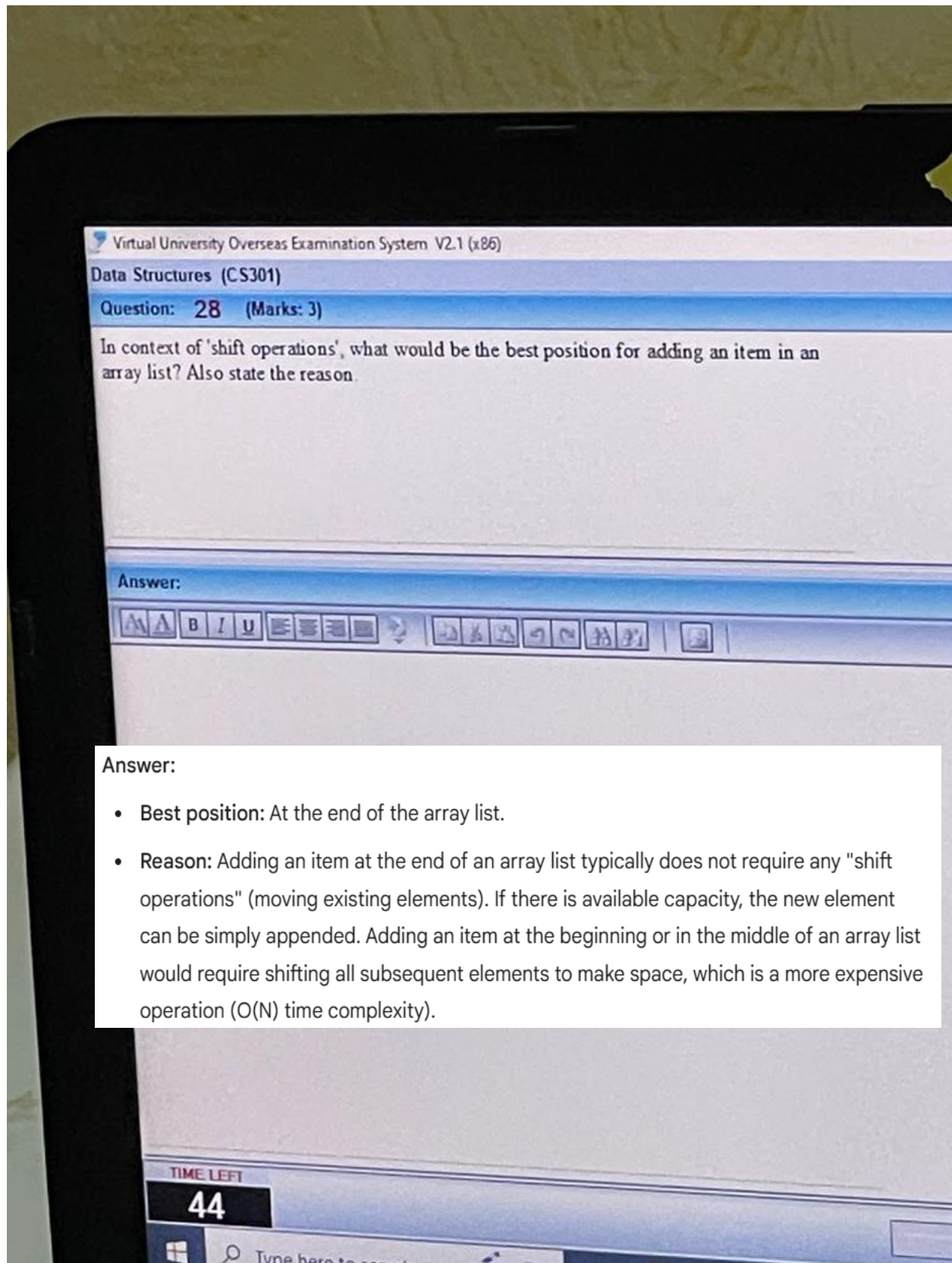
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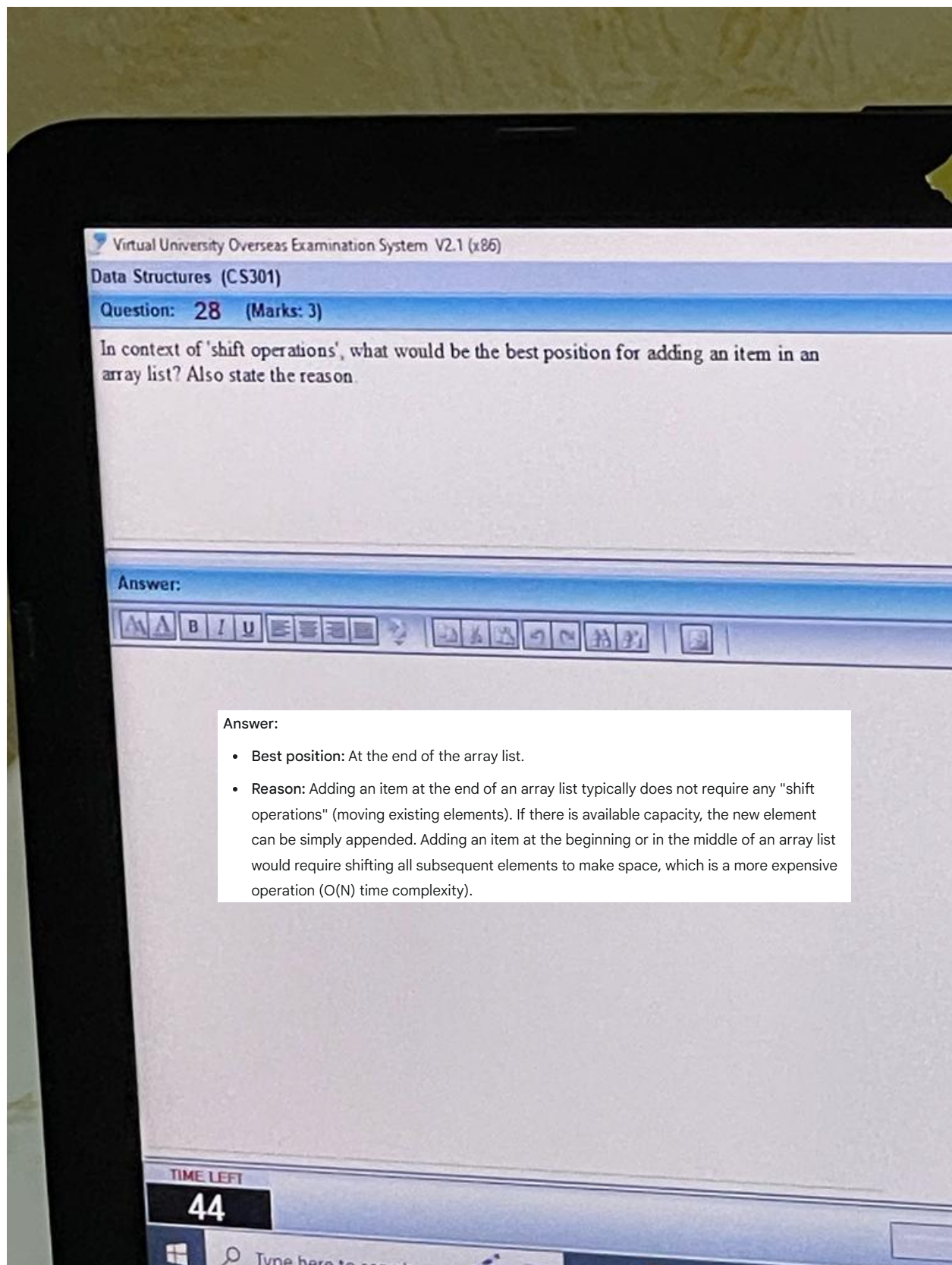
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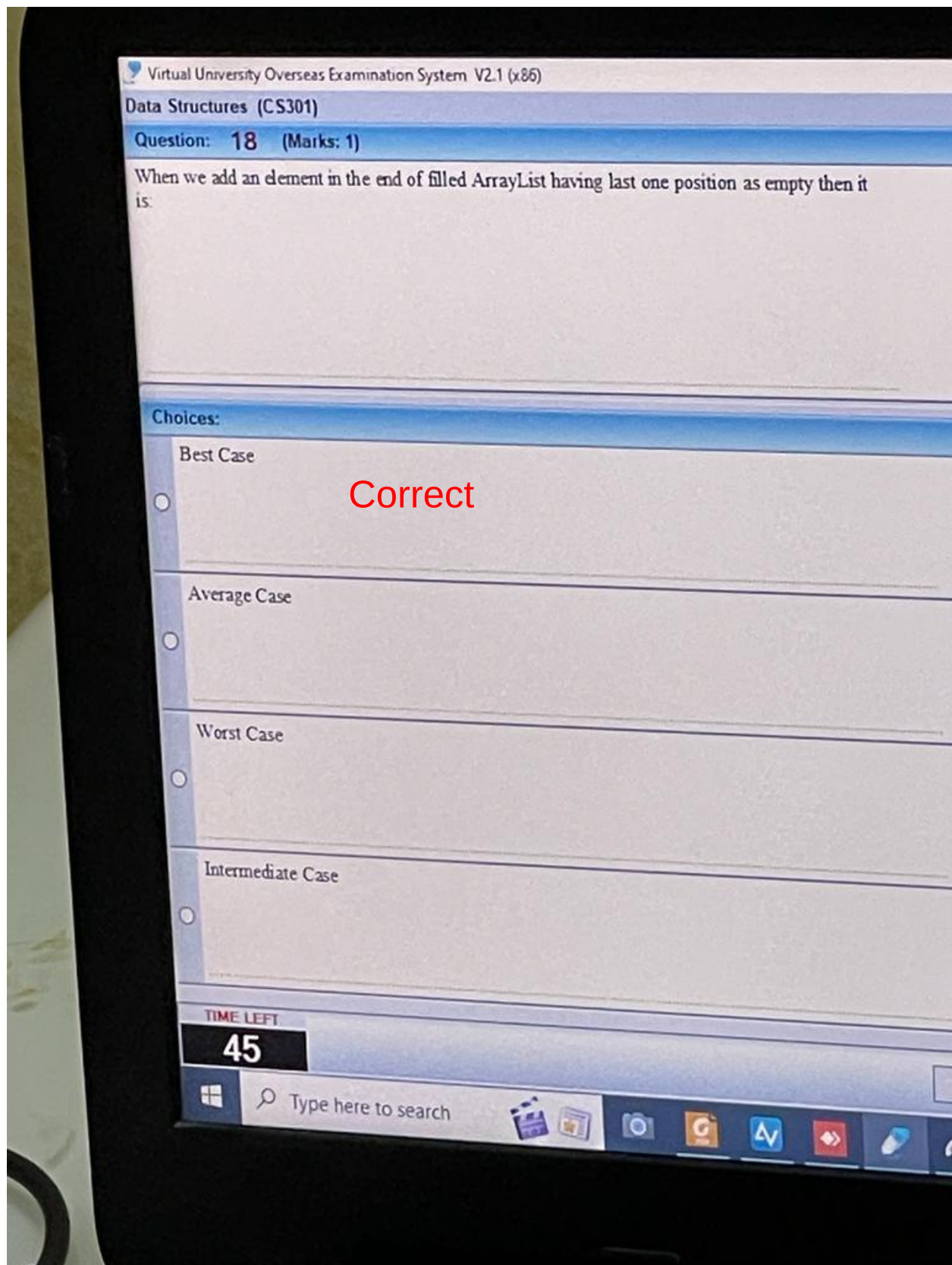
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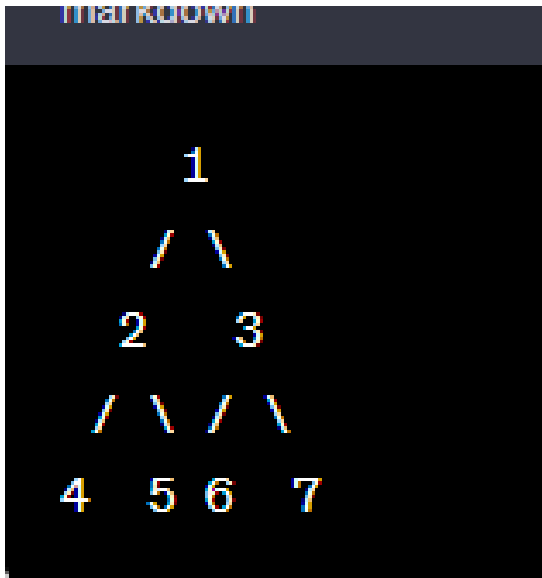
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Solution

Q31 part a

Next is length() method as the size of heap is also that of the queue. Therefore, length() is internally calling the getSize() method of heap.

Part b



Q32

Yes

Those recursive routines were very small, 3 to 4 lines of code and they could be employed to traverse a tree of any size. We also traversed BST in inorder to retrieve the information in sorted order. We employed stacks in recursive implementations. Although, recursive routines are of few lines but when recursion is in action, recursive stack is formed that contains the function calls. We also explicitly used stack for inorder non-recursive traversal.

Example of inorder traversal

```
class Node:
    def __init__(self, key):
        self.left = None
        self.right = None
```


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```
self.val = key
```

```
def inorder_traversal(root):
```

```
    if root:
```

```
        # Traverse the left subtree
```

```
        inorder_traversal(root.left)
```

```
        # Visit the root node
```

```
        print(root.val, end=" ")
```

```
        # Traverse the right subtree
```

```
        inorder_traversal(root.right)
```

```
# Example usage:
```

```
# Create a sample binary tree
```

```
root = Node(1)
```

```
root.left = Node(2)
```

```
root.right = Node(3)
```

```
root.left.left = Node(4)
```

```
root.left.right = Node(5)
```

```
# Perform inorder traversal
```

```
print("Inorder Traversal:")
```

```
inorder_traversal(root)
```

```
Q29
```

```
isEmpty()
```

```
front()
```

```
dequeue()
```

```
enqueue(X)
```

```
Q28
```

```
Add method
```

```
Reason
```

```
Suppose if the size of the
```

list is 10000 or 20000, we have to do the shift operation for all of these 10000 or 20000 elements. Normally, it is done by shifting of elements with the use of a for loop. This operation takes much time of the CPU and thus it is not a good practice to add an element at the beginning of a list. On the other hand, if we add an element at the end of the list, it can be done by carrying out 'no shift operation'. It is the best case of adding an element to the list. However, normally we may have to move half of the elements. The usage of add method is the matter warranting special care at the time of implementation of the list in our program. To provide the interface of the list, we just define these metho

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CS301

Q1 C	Q9	(17)	(25) A
Q2 C	Q10	(18) A	(26) A
Q3 D	Q11	(19) B	(27)
Q4 D	Q12	(20) A	
Q5	Q13	(21) A	
Q6 B	(14)	(22) B	
Q7 B	(15) A	(23) C	
Q8	(16) D	(24) A	